

Plant traits shape global spatiotemporal variations in photosynthetic efficiency

Received: 14 September 2024

Accepted: 26 February 2025

Published online: 25 March 2025



Yulin Yan^{1,2}✉, Bolun Li³, Benjamin Dechant^{4,5}, Mingzhu Xu²,
Xiangzhong Luo⁶, Sai Qu⁶, Guofang Miao², Jiye Leng⁷, Rong Shang²,
Lei Shu², Chongya Jiang⁸, Han Wang⁹, Sujong Jeong¹⁰, Youngryel Ryu¹¹ &
Jing M. Chen^{12,7}✉

Photosynthetic efficiency (PE) quantifies the fraction of absorbed light used in photochemistry to produce chemical energy during photosynthesis and is essential for understanding ecosystem productivity and the global carbon cycle, particularly under conditions of vegetation stress. However, nearly 60% of the global spatiotemporal variance in terrestrial PE remains unexplained. Here we integrate remote sensing and eco-evolutionary optimality theory to derive key plant traits, alongside explainable machine learning and global eddy covariance observations, to uncover the drivers of daily PE variations. Incorporating plant traits into our model increases the explained daily PE variance from 36% to 80% for C_3 vegetation and from 54% to 84% for C_4 vegetation compared with using climate data alone. Key plant traits—including chlorophyll content, leaf longevity and leaf mass per area—consistently emerge as important factors across global biomes and temporal scales. Water availability and light conditions are also critical in regulating PE, underscoring the need for an integrative approach that combines plant traits with climatic factors. Overall, our findings demonstrate the potential of remote sensing and eco-evolutionary optimality theory to capture principal PE drivers, offering valuable tools for more accurately predicting ecosystem productivity and improving Earth system models under climate change.

Plant photosynthesis is the fundamental process that sustains ecosystem function, drives the carbon cycle, regulates atmospheric carbon dioxide (CO_2) levels and influences global climate dynamics^{1,2}. Photosynthetic efficiency (PE) or light use efficiency, defined as the ratio of gross primary productivity (GPP) to absorbed photosynthetically active radiation (APAR), is a crucial metric for assessing how effectively ecosystems use the energy from absorbed sunlight for photosynthesis³. Understanding and optimizing PE is essential for improving crop yields, enhancing

carbon sequestration and developing sustainable strategies to mitigate climate change^{4–8}. However, around 60% of the spatiotemporal variability in ecosystem-scale PE remains unexplained^{9,10}, underscoring the urgent need to better understand the factors that control this essential process.

Integrating plant traits offers a promising path to advancing our knowledge of ecosystem PE. Although primary climatic factors such as cloudiness index (CI_{cloud}), temperature, soil moisture and vapour pressure deficit (VPD) are crucial^{11–13} and supported by well-documented

¹Geography Postdoctoral Program, Fujian Normal University, Fuzhou, China. ²Key Laboratory of Humid Subtropical Eco-Geographical Process (Ministry of Education), Fujian Normal University, Fuzhou, China. ³School of Geographical Sciences, Nanjing University of Information Science and Technology, Nanjing, China. ⁴German Centre for Integrative Biodiversity Research, Leipzig, Germany. ⁵Leipzig University, Leipzig, Germany. ⁶Department of Geography, National University of Singapore, Singapore, Singapore. ⁷Department of Geography and Planning, University of Toronto, Toronto, Ontario, Canada. ⁸College of Agriculture, Nanjing Agricultural University, Nanjing, China. ⁹Department of Earth System Science, Tsinghua University, Beijing, China. ¹⁰Department of Environmental Planning, Seoul National University, Seoul, South Korea. ¹¹Department of Landscape Architecture and Rural Systems Engineering, Seoul National University, Seoul, South Korea. ✉e-mail: yulin.yan@outlook.com; jing.chen@utoronto.ca

physiological mechanisms^{9,10,13–20}, they collectively explain less than half of the spatiotemporal variation in ecosystem-scale PE^{9,10}. In contrast, plant traits—including morphological, anatomical and biochemical features—mechanistically determine how plants respond to environmental cues and regulate photosynthetic machinery^{21–24}. For example, leaf chlorophyll content (Chl) and the maximum carboxylation rate (V_{cmax}) of Rubisco are central to photosynthesis due to their direct roles in light absorption and the Calvin–Benson cycle, which assimilates CO_2 into carbohydrate^{2,15,25}. Additionally, leaf longevity (LL) is one fundamental trait in the leaf economic spectrum that can serve as an indicator of other photosynthetic traits^{22,26}. LL and leaf mass per area (LMA) reflect plants' strategies to optimize carbon gain relative to structural investment^{26–29}. Canopy structure, including leaf area index (LAI) and clumping index (CI_{clump}), directly impacts light absorption through radiative transfer³⁰. Despite their importance, these traits remain underexplored for estimating ecosystem-scale PE.

Linking plant traits to macroscale PE remains challenging due to the scarcity of spatiotemporally synchronized trait data and the difficulty of upscaling in situ measurements to align with eddy covariance flux observations^{31,32}. Although efforts have been made to link plant traits with GPP^{31,33–35}, debate persists about their effectiveness in predicting ecosystem functional properties^{36–38}, which serve as unified metrics that describe ecosystem processes and responses in a comparable way, such as PE^{4,5}. This raises several critical questions⁴: To what extent do plant traits capture the substantial unexplained (60%) variability of ecosystem PE across global biomes? How do canopy structure and climatic conditions influence the relationship between plant traits and ecosystem PE?

Here we integrate remote sensing (RS) and eco-evolutionary optimality (EEO) theory to address the challenge of unavailable in situ plant trait measurements at eddy covariance towers. EEO theory hypothesizes that plant traits evolve to optimize characteristics that balance competing needs, making them the best fit for their environment²⁶. By leveraging data from the global eddy covariance network (120 sites, FLUXNET2015), we aim to understand the contributions of plant traits and climate to the variance in daily ecosystem PE during the growing season across various terrestrial ecosystems. Using explainable machine learning methods, we seek to elucidate the mechanisms driving ecosystem-scale PE in the terrestrial biosphere.

Results

Plant traits and terrestrial ecosystem PE

To quantify the extent to which plant traits contribute to the variation in ecosystem PE across global biomes, we employed three scenarios: climatic factors only, plant traits only and a combination of all factors. Our results show that climatic factors alone account for merely 36% of the spatiotemporal variance in daily PE (Fig. 1a,d), consistent with current understanding of environmental controls^{9,10}. In contrast, plant traits alone explain a substantial 67% of the variance (Fig. 1b,e), with static (growing season mean) plant traits contributing 36% (Extended Data Fig. 1). Although EEO-based traits reflect long-term climate acclimations, explicitly integrating them into models markedly enhances predictive performance, as machine learning models alone cannot fully capture these climate-driven transformations. By incorporating both daily climate data and EEO-based traits, we account for both the immediate, direct effects of climate and the indirect, long-term effects that shape ecosystem PE. The combination of climatic factors and plant traits provides the highest explanatory power (coefficient of determination (R^2), 0.80) and the lowest uncertainty (root mean squared error (RMSE), 30.5%) (Fig. 1c,f). These findings highlight the importance of an integrative approach to enhance our understanding of ecosystem PE. The influence of plant traits on PE persists across both weekly and biweekly timescales (Supplementary Fig. 13 and Supplementary Information). The robustness of these conclusions is further validated by analyses conducted exclusively on C_4 vegetation, demonstrating an

improvement in R^2 from 54% to 84% compared with using climate data alone (Extended Data Fig. 2).

RS and EEO traits' roles in ecosystem PE

Having confirmed the critical role of plant traits in explaining ecosystem PE, we next identified specific traits that contribute additional explanatory power beyond climate variables. To achieve this, we assessed the independent contributions of RS-based traits (Chl, CI_{clump} and LAI) and EEO-based traits (LL, LMA and V_{cmax} normalized at 25 °C ($V_{\text{cmax}25}$)) by evaluating how each trait improves model performance when added to a baseline model using only climate variables (Fig. 2).

Integrating plant traits derived from either RS or EEO considerably enhances the model's ability to explain the variance in ecosystem PE, while also reducing estimation errors across multiple metrics (Fig. 2). Among all the traits, LMA and LL were particularly impactful across R^2 , RMSE, slope and mean absolute percentage error metrics. However, Chl emerged as the most influential trait overall, as indicated by its Shapley additive explanations (SHAP) value, which reflects its additive contribution to the model's predictions rather than the model's quality compared with observations (Fig. 1e,f). The combined addition of all EEO traits outperformed the collective RS traits, and the integration of all plant traits (RS and EEO) delivered the best overall performance, underscoring the complementary strengths of these traits. These results highlight the efficacy of both RS and EEO methods in capturing the primary drivers of ecosystem PE, providing robust tools for tracking ecosystem productivity and detecting functional changes in response to climate change.

Conversely, to assess the additional contributions of climatic variables, we started with a baseline of plant traits only and subsequently incorporated climatic factors (Extended Data Fig. 3). Our findings confirm that while plant traits are the primary drivers of ecosystem PE, factors such as water availability and light conditions (CI_{cloud}) also play crucial roles in mediating carbon fluxes (Fig. 1a,d and Extended Data Fig. 3). Specifically, although the evaporative stress index (ESI), defined as the ratio of actual to potential evapotranspiration, showed a moderate effect, only VPD and CI_{cloud} notably improved the explanatory power (Extended Data Fig. 3). The limited enhancement from adding climate variables probably stems from the fact that EEO-derived plant traits have already partially incorporated climatic information through the EEO theory, such as the temperature-dependent maximum light-adapted quantum yield of photosystem II (Methods, equation (15)). These findings reinforce the necessity of an integrative approach that combines plant traits and climatology to fully understand ecosystem PE. The robustness of these conclusions is further supported by analyses using alternative datasets (for example, the MERIS Chl product; Extended Data Fig. 4).

Plant traits explain PE across global biomes

To validate the robustness of our key findings from the general model, which was trained on the entire dataset (Figs. 1 and 2), we extended our analysis by training and evaluating the Extreme Gradient Boosting (XGBoost) machine learning model separately for each plant functional type (PFT). This analysis was conducted under two scenarios: one considering only climatic factors, and the other incorporating both climatic factors and plant traits, with SHAP analysis providing further interpretive insights. Despite disparities in subset data sizes, the XGBoost model consistently performed relatively well in predicting ecosystem PE across global major PFTs (Fig. 3 and Extended Data Figs. 5 and 6). The model achieved a mean R^2 ranging from 0.64 to 0.81 and RMSE from 22% to 49%, with slopes ranging from 0.85 to 1.03. Cropland exhibited the greatest improvement when plant trait information was incorporated (Fig. 3), followed by grassland, deciduous broadleaf forest and evergreen needleleaf forest. However, the limited data size for certain PFTs, such as savanna (validation $n = 240$), probably influenced model performance (Extended Data Fig. 6). Moreover, we observed

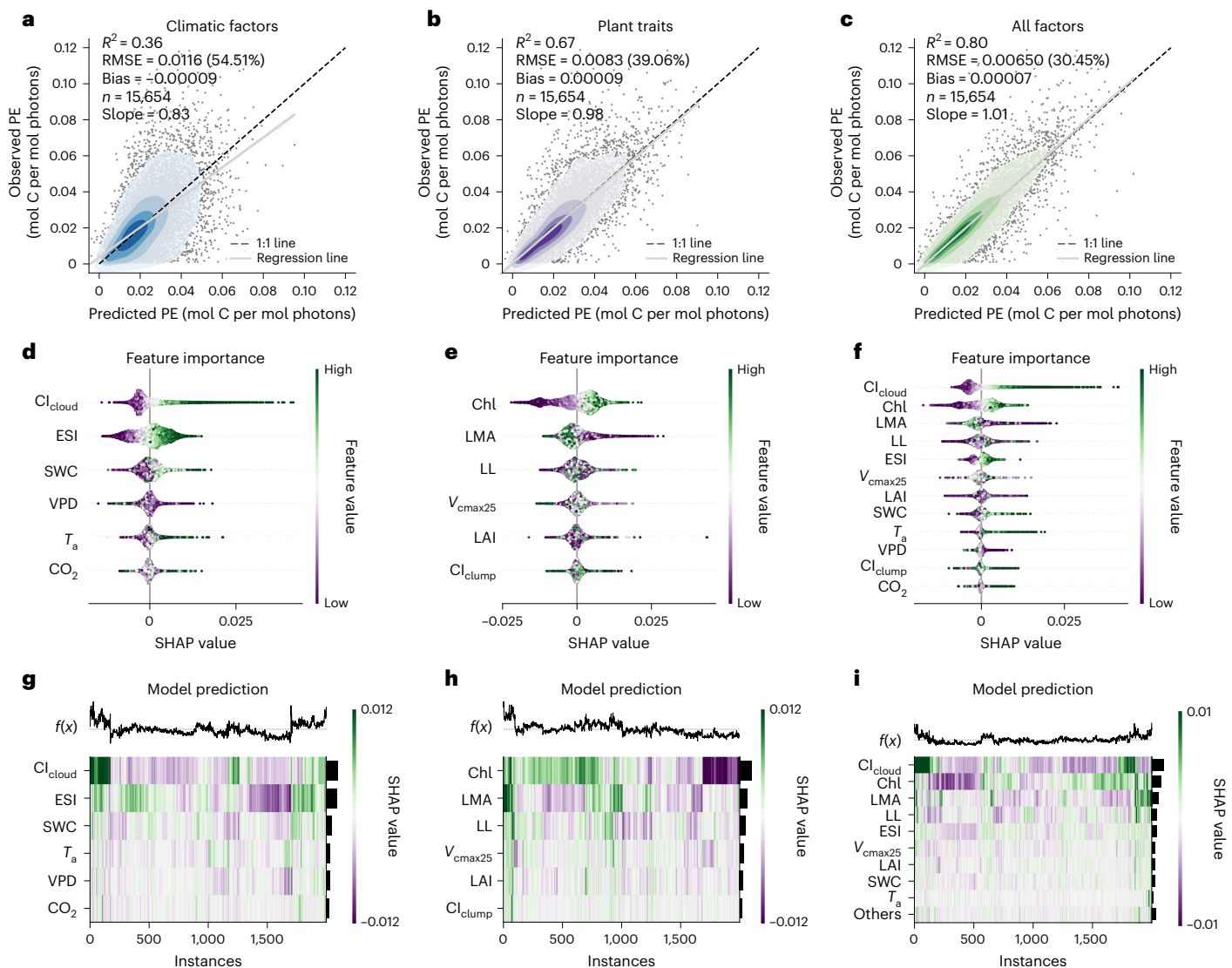


Fig. 1 | Explanatory power of various factors in estimating daily variation in ecosystem-scale PE for C_3 vegetation. a–i, Results for climatic factors only (a,d,g), leaf traits and canopy structure (b,e,h), and leaf traits, climatic factors and canopy structure (c,f,i). In a–c, the shaded areas indicate Gaussian kernel density estimations. In d–f, the contribution (SHAP value) of each feature is

ranked in descending order of importance, along with their relative feature values. In g–i, the heat maps show clustering of 2,000 random instances by their local explanation embedding. The columns represent instances, and the rows represent features' SHAP values for each explanatory variable, where $f(x)$ denotes the model-predicted PE for each instance.

that the contribution of plant traits (for example, Chl, LL and V_{cmax25}), water availability metrics (ESI, soil water content (SWC) and VPD) and diffuse radiation remained solid in shaping ecosystem PE, albeit with varying feature importance across different PFTs (Extended Data Fig. 5). These findings highlight the crucial role of plant traits in influencing ecosystem PE across global biomes.

Main effects and underlying mechanisms

Following the identification of the primary drivers, we investigated how plant traits and climate influence ecosystem PE. We present the detailed marginal contributions of these factors to elucidate the underlying mechanisms (Fig. 4). Our findings reveal notable nonlinear contributions from both plant traits and climate, indicating intricate interactions and feedback across terrestrial ecosystems. For instance, Cl_{cloud} has a dominant influence on PE, exerting an exponential impact that surpasses those of all other variables (Figs. 1f and 4a). Chl also plays a pivotal role, especially at lower levels. At levels below approximately $20 \mu g cm^{-2}$, Chl constrains PE, probably through insufficient

light harvesting, leading to negative contributions. However, as Chl increases, PE shows notable enhancement, although the rate of increase tends to diminish (Fig. 4b). LL presents a more complex picture: while SHAP values alone reveal no distinct pattern, our results indicate that shorter LL tends to be associated with higher PE (Fig. 4c and Supplementary Fig. 4d). Similarly, lower LMA ($<100 g m^{-2}$) is associated with a positive impact on PE (Fig. 4d), consistent with the established positive LMA–LL relationship²³; this suggests that a higher turnover rate of canopy leaves, reflected in a greater fraction of younger leaves, enhances PE.

Moreover, an LAI below -3 enhances PE, while further increases reduce it (Fig. 4f), indicating that although a denser canopy captures more light, it converts this light into productivity less efficiently. These nonlinear contributions such as those of V_{cmax25} and LAI are also influenced by the uncertainties in remotely sensed fraction of APAR (fPAR) and LAI (Supplementary Figs. 11 and 12 and Supplementary Section 1.5). Furthermore, ESI, quantified by the ET/PET index, exhibits a positive impact on PE; this impact becomes particularly pronounced

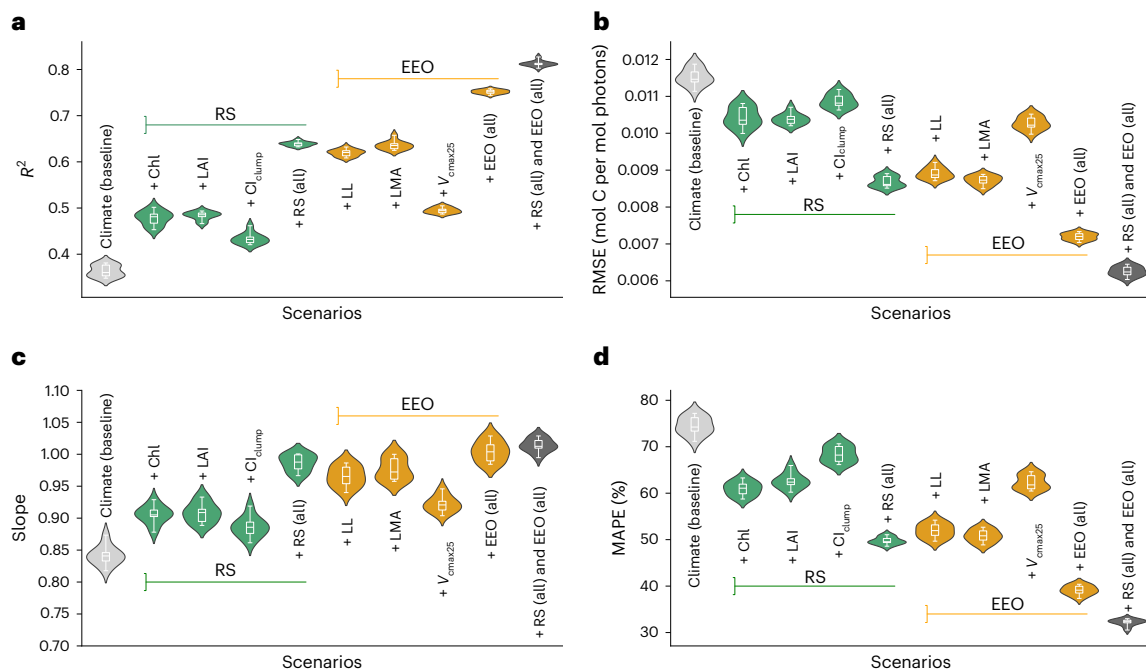


Fig. 2 | Performance in estimating daily ecosystem-scale PE with additional contributions of plant traits for C_3 vegetation, including RS traits (Chl, LAI and Cl_{clump}) and EEO traits (LL, LMA and V_{cmax25}) incorporated into the baseline of the climate-only model (Cl_{cloud} , ESI, SWC, VPD, T_a and CO_2). **a**, Variation explained. **b**, RMSE. **c**, Slope of the linear regression lines against FLUXNET2015 observed PE. **d**, Mean absolute percentage error (MAPE). The box plots show the results from k -fold cross-validation ($n = 10$), where each replicate represents a distinct subset of data ($n = 7,827$) derived from FLUXNET2015 sites. The box

plots are overlaid with violin plots (coloured areas). The width of each violin corresponds to the frequency of observations at specific values. In each box plot, the bounds of the box represent the interquartile range (25th to 75th percentile), the line within the box indicates the median, and the whiskers extend to the minimum and maximum values within 1.5 times the interquartile range. The control is the baseline climate-only model, and comparisons are made to models incorporating RS and EEO traits. Error metrics were derived exclusively from all replicates.

when the ESI exceeds -0.6 , marking a transition from water-limited to water-favourable conditions, thereby enhancing PE (Fig. 4g). SWC positively influences PE up to approximately 30%, after which its effect plateaus (Fig. 4h). Conversely, higher VPD negatively impacts PE (Fig. 4i). Temperature positively contributes to PE around 20 – 35 °C. Cl_{clump} and CO_2 levels have minimal direct effects on PE within the observed range. However, it is important to note that the FLUXNET2015 dataset used in this study primarily reflects short-term CO_2 effects, which may limit the generalizability of these findings over longer timescales.

Interaction effects and underlying mechanisms

To investigate how potential interaction effects shape the nonlinear contributions of plant traits to PE (Fig. 4), we analysed SHAP interaction values (Extended Data Fig. 7). While widespread coupling was observed among leaf traits, canopy structure and climate variables (Extended Data Fig. 7), our analysis focused on variables with Spearman's $\rho < 0.75$ (Supplementary Fig. 2d). We selected this slightly higher threshold to balance reducing multicollinearity with the need to retain critical, closely related variables such as Chl, LAI and V_{cmax25} .

Our findings uncover complex trade-offs among light distribution, harvesting efficiency and photosynthetic capacity, driven by interactions between canopy structure, biochemical traits and climate. For instance, elevated Chl amplifies the positive impact of Cl_{cloud} on PE, whereas lower Cl_{cloud} levels diminish this effect (Extended Data Figs. 4c and 8a). Notable interactions also emerge among LAI, Chl and V_{cmax25} , with dense canopies ($LAI > 4$) reducing the benefits of increased Chl on PE, probably due to light limitations (Extended Data Fig. 8b). This negative interaction is consistent across alternative Chl products (MERIS; Extended Data Fig. 4d).

Additionally, higher LMA mitigates the positive effect of Chl, suggesting a trade-off between structural investment and light use

(Extended Data Fig. 8c). Conversely, LMA enhances the positive impact of V_{cmax25} , indicating that thicker leaves may better exploit increased photosynthetic capacity (Extended Data Fig. 8d). However, the benefits of high V_{cmax25} are reduced in dense canopies (Extended Data Fig. 8e). Lastly, shorter LL combined with higher V_{cmax25} boosts PE, although this advantage wanes with longer LL, reflecting potential reductions in ecosystem productivity (Extended Data Fig. 8f). These findings reveal the intricate balance between light-harvesting and biochemical traits, underscoring the need for further research to disentangle these interactions and fully understand their implications for ecosystem-scale PE.

Discussion

In this study, we present a data-driven framework that notably enhances our understanding of the key drivers behind ecosystem PE in the terrestrial biospheres (Figs. 1–4). Our analyses identify plant traits as valuable contributors to ecosystem PE across diverse global biomes and various timescales. Our findings reveal that the combination of plant traits and climate variables accounts for 80% (C_3) to 84% (C_4) of the variations in PE. This integrative approach provides new insights into the intricate interplay between biological and environmental factors that drive photosynthetic processes at the ecosystem level.

Our findings highlight the critical role of plant traits in modulating ecosystem processes and indicate that relying solely on climatic variables or plant traits alone is inadequate to fully explain the intricate variations in these processes (Figs. 1 and 2). For instance, previous studies have shown that the effect of VPD on ecosystem PE varies across biomes^{39,40}. This complexity may be due to its interaction with temperature^{10,41} or decoupling from soil moisture^{42,43}, or potentially unaccounted-for plant traits. Our results also reveal that the complex contribution of LL to ecosystem PE becomes apparent when we account for the interplay with

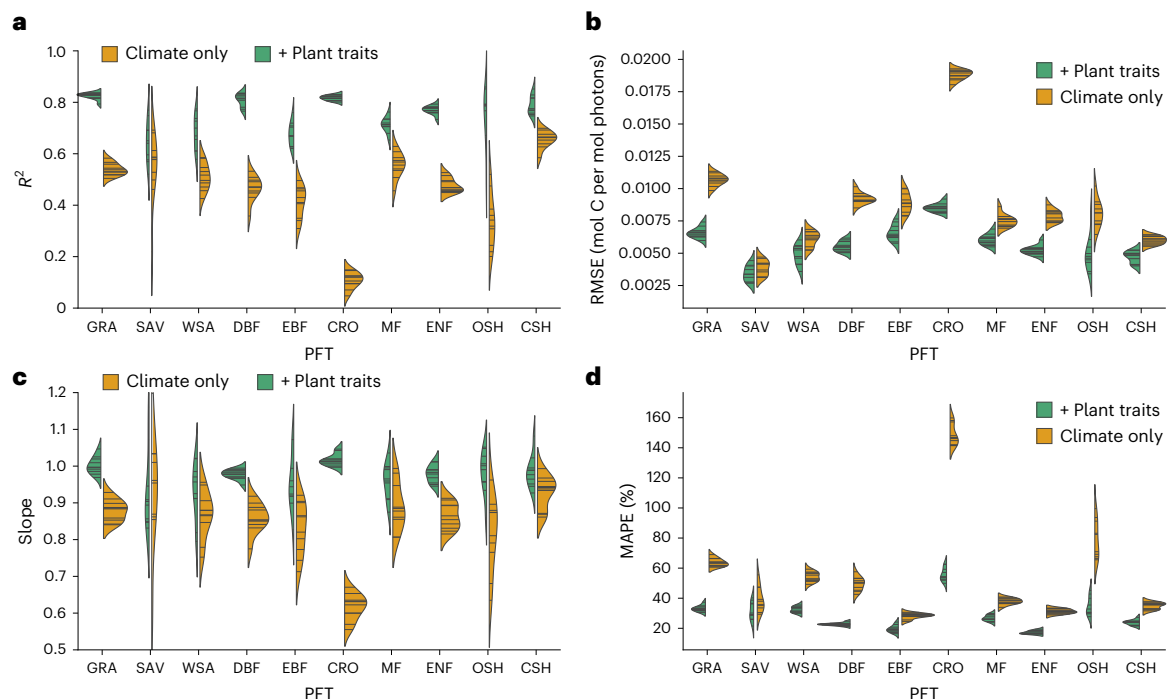


Fig. 3 | Performance in estimating daily ecosystem-scale PE for C_3 vegetation across global major PFTs, presented in split violin plots. a, Variation explained. **b**, RMSE. **c**, Slope of the linear regression lines against FLUXNET2015 inferred PE. **d**, MAPE. CRO, cropland; CSH, closed shrubland; DBF, deciduous broadleaf

forest; EBF, evergreen broadleaf forest; ENF, evergreen needleleaf forest; GRA, grassland; MF, mixed forest; WSA, woody savannah; SAV, savannah; OSH, open shrubland. The violin plots represent the results from k -fold cross-validation ($n = 10$).

biochemical traits such as $V_{\text{cmax}25}$ (Extended Data Fig. 8f). Moreover, our results show that climatic variables influence photosynthesis indirectly through interactions with leaf traits. For instance, an increase in diffuse radiation enhances light availability on shaded leaves⁴⁴, which, when paired with high Chl, leads to more efficient light use, thereby boosting canopy photosynthesis^{45–48}. This interaction between diffuse radiation and Chl vividly illustrates the necessity of considering both climatic factors and plant traits in tandem in ecosystem modelling.

Canopy structure is crucial for unravelling the complexities of ecosystem photosynthesis^{49,50}. Canopy structure influences the light environment via radiative transfer, directly affecting light absorption and photosynthetic activity³⁰. In dense canopies with high LAI, light penetration is notably reduced, particularly in the lower canopy layers, as described by the Beer–Lambert law⁵¹. This leads to light-limited photosynthesis in shaded leaves, diminishing the overall contribution of chlorophyll pigments (Extended Data Fig. 8b). The interplay between diffuse radiation and chlorophyll (Extended Data Figs. 4c and 8a) highlights a critical point: while plants have evolved to optimize light capture through vertical gradients in leaf nitrogen and chlorophyll^{52,53}, our findings raise questions about how these gradients interact with diffuse light. Future studies incorporating multi-layer canopy modelling and higher-resolution data are needed to explore whether these gradients represent the most effective strategy under varying light regimes. This observation also highlights a diminishing return on vegetation's investment in chlorophyll, as seen in Fig. 4b, where the balance between productivity (for example, higher LAI) and PE becomes evident. The trade-off indicates that while vegetation continues to aim for high productivity, it does so through different strategies, either by maximizing LAI at the cost of PE or by optimizing PE with lower LAI (Fig. 4f).

Our study advances understanding of ecosystem photosynthesis, providing a fresh view for next-generation Earth system models. Currently, key plant traits such as $V_{\text{cmax}25}$, LAI and Cl_{clump} are well integrated into process-based biophysical models, due to their central roles in either the FvCB model or canopy radiative transfer and energy balance

processes^{54,55}. Chl is commonly used to empirically parameterize $V_{\text{cmax}25}$ (ref. 56), though future work could explore its direct integration by modifying parameters related to light absorption and energy conversion (for example, J_{max}), especially under fluctuating light conditions. LL can also constrain $V_{\text{cmax}25}$ as a function of relative age to indicate the leaf turnover rate, although this remains an empirical approach⁵⁷. Similarly, LMA could be explored as a constraint on other traits, such as LAI and $V_{\text{cmax}25}$, given its role within a broader network of interconnected plant traits that collectively govern plant performance^{58,59}. To refine photosynthesis models, evaluations should focus on PE rather than solely on GPP, which is often dominated by canopy structure. Therefore, while it is crucial to incorporate key plant traits into dynamic global vegetation models to better capture phenomena such as extreme climate events and interannual GPP variations, this effort must carefully balance simplicity, realism and uncertainty^{6,60}. However, these efforts are hindered by the limited availability of global plant trait data. We propose an interdisciplinary approach that leverages climate data, RS (for example, Chl, LAI and Cl_{clump})^{61,62}, EEO principles (for example, LL, LMA and $V_{\text{cmax}25}$)^{26,63} and global plant trait databases^{22,64}. This integration is essential for a comprehensive understanding of how ecosystem-scale photosynthesis responds to environmental changes. Leaf nitrogen and phosphorus are critical yet underrepresented traits for understanding ecosystem PE, primarily due to the scarcity of continuous, broad-scale data⁶⁵. By integrating empirically derived leaf nitrogen and phosphorus content into our models, we notably improved their explanatory power and overall performance (Supplementary Fig. 1 and Supplementary Information). This enhancement underscores the critical role these nutrients play in regulating ecosystem productivity^{66,67}. However, the strong correlations between nitrogen, phosphorus and $V_{\text{cmax}25}$ suggest potential confounding effects, making the interpretation challenging. Besides, the plastic allocation of nitrogen to Rubisco, LMA and Chl could introduce considerable uncertainties in empirically derived leaf nitrogen estimates^{68,69}. Consequently, our study should be viewed as an initial exploration of these complex relationships. Reassessing these

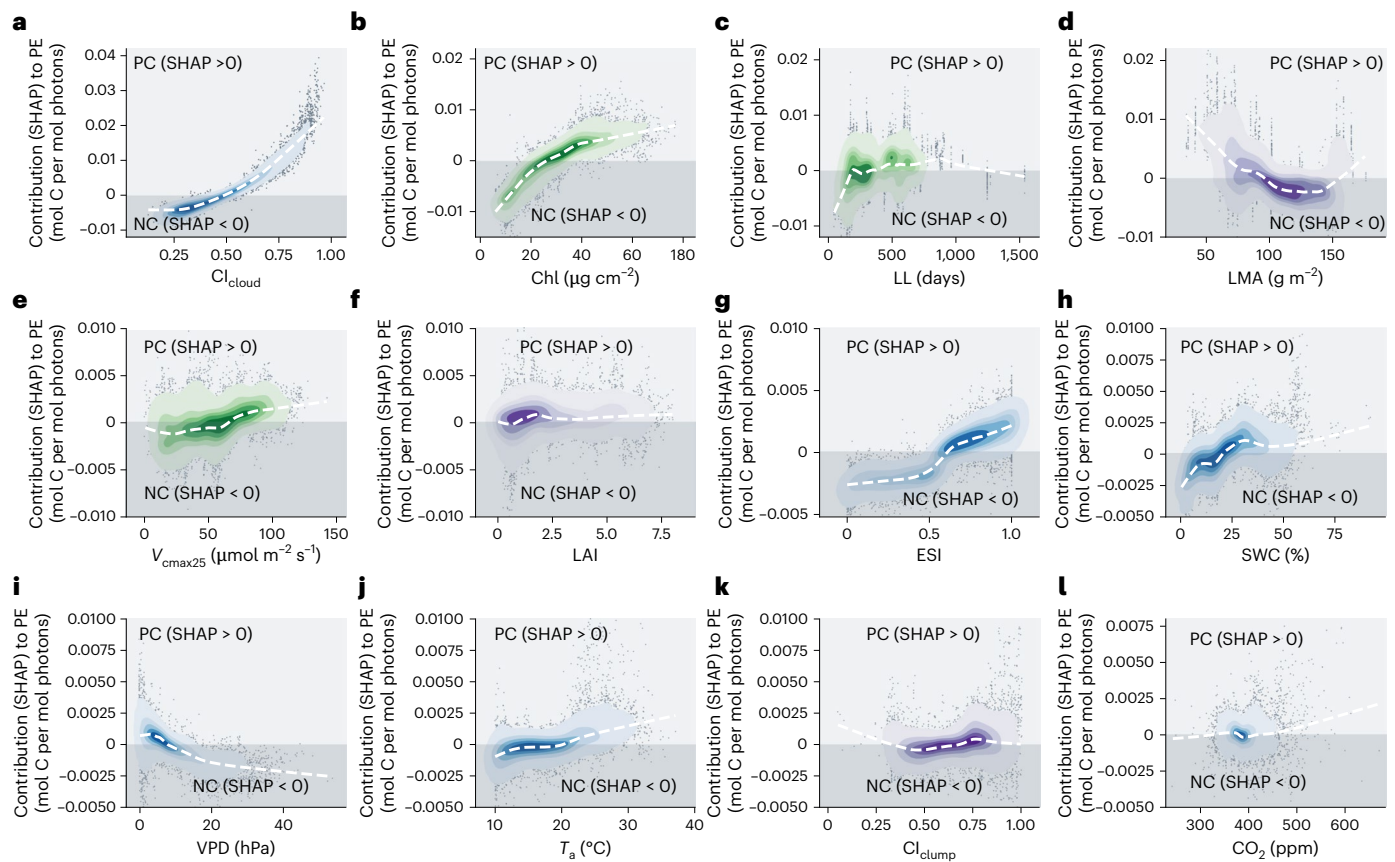


Fig. 4 | Contributions of leaf traits, canopy structure and climate factors to daily ecosystem-scale PE for C_3 vegetation. a–l, Blue represents climate variables, green denotes biochemical plant traits and purple indicates structural plant traits. Positive SHAP values signify a positive influence on PE, while negative values indicate a negative effect. The colour-shaded areas reflect

Gaussian kernel density estimations. The white dashed lines represent the non-parametric LOWESS trend, which smooths the data to reveal underlying patterns without assuming a specific functional form. PC, positive contribution; NC, negative contribution.

dynamics will be crucial when global-scale foliar trait maps, enabled by hyperspectral satellite data, become available.

While EEO models offer valuable insights into capturing the spatial variation among sites in traits such as LL and LMA²⁶, they rely on the assumption of a steady state relative to long-term (for example, growing season) environmental conditions. This reliance makes it challenging to predict seasonal dynamics accurately. However, this limitation does not extend to $V_{\text{cmax}25}$ (ref. 63). Additionally, EEO methods are predominantly driven by climate, making them less direct than approaches that consider light absorption features linked to leaf chemical composition. Hyperspectral RS has shown notable promise in mapping global plant traits, with current operational missions (for example, EnMAP and GF-5) and upcoming missions (for example, SBG and CHIME)⁷⁰ offering enhanced spatial–temporal resolution and accuracy, particularly for traits such as Chl, leaf nitrogen and LMA. Moreover, the integration of solar-induced fluorescence measurements has emerged as a powerful tool to directly assess plant photosynthetic activity, providing complementary insights into vegetation function⁷¹. These advancements could greatly improve our ability to capture the spatial–temporal patterns of vegetation characteristics, including PE, leading to a more comprehensive understanding of ecosystem responses to climate change. However, accurately incorporating plant traits and representing photosynthesis in Earth system models—whether through empirical or mechanistic approaches—remains a notable challenge and a crucial objective for future research^{6,60,72,73}. A hybrid modelling approach that integrates process-based models with the adaptability of data-driven machine learning offers a promising path forward⁷⁴.

Methods

Data processing and ecosystem-scale PE

We used the FLUXNET2015 Tier 1 daily dataset, which covers the eddy covariance flux measurements of carbon, water vapour and energy exchange between ecosystems and the atmosphere⁷⁵. We used daily GPP data (GPP_NT_VUT_REF) based on the nighttime decomposition method⁷⁶. Daily gap-filled meteorological records include air temperature (T_{a}), incoming surface shortwave radiation (SW_{IN_F}), potential shortwave radiation ($\text{SW}_{\text{IN}_{\text{POT}}}$), surface pressure (PA_F), VPD (VPD_F), CO_2 ($\text{CO2}_F_{\text{MDS}}$) and shallow SWC ($\text{SWC}_F_{\text{MDS}_1}$). All data were filtered using a data quality control threshold set at >0.8 . We also excluded negative GPP, VPD, SWC and CO_2 data. We calculated Cl_{cloud} as the alternative for the fraction of diffuse solar radiation⁹:

$$\text{Cl}_{\text{cloud}} = 1 - \left(\frac{\text{SW}_{\text{IN}_F}}{\text{SW}_{\text{IN}_{\text{POT}}}} \right) \quad (1)$$

where SW_{IN_F} is incoming surface shortwave radiation and $\text{SW}_{\text{IN}_{\text{POT}}}$ is potential shortwave radiation. Ecosystem PE is defined as GPP normalized by APAR:

$$\text{PE} = \frac{\text{GPP}}{\text{fPAR} \times \text{PPFD}} \quad (2)$$

where fPAR (unitless) is the fraction of APAR (mol photons per m^2 per day) obtained from MODIS MOD15A2H (eight-day, 18 February 2000 to 3 July 2002) and MCD15A3H (four-day, 4 July 2002 onwards) v.6.1

products, both at a $1.0 \times 1.0 \text{ km}^2$ resolution. The original data were quality controlled (FparLai_QC and FparExtra_QC) and smoothed with a Savitzky–Golay filter. fPAR was interpolated to daily scale using the Akima interpolator for gaps smaller than 50%⁷⁷. Moreover, to assess uncertainties stemming from the choice of fPAR products, we compared results with additional quality-controlled 1-km GEOV2 fPAR (ten-day, 1 January 2000 onwards)⁷⁸ (Supplementary Fig. 12). PPFD (mol photons per m^2 per day) is the photosynthetic photon flux density estimated as a function of SW_IN_F (W m^{-2}) using a multiplier of $2.04 \mu\text{mol J}^{-1}$ (ref. 79). We calculated PE at three temporal scales: daily, 7-day mean and 14-day mean, using consecutive gap-free and non-overlapping rolling windows accordingly to prevent temporal autocorrelation (Supplementary Fig. 13b,c). PE was further constrained to values greater than 0.001 mol C per mol photons and less than 0.12 mol C per mol photons⁹.

Our final dataset included 120 sites and 10 PFTs defined by the IGBP (Supplementary Table 1), including 6 EBF sites, 16 DBF sites, 7 MF sites, 28 ENF sites, 6 WSA sites, 6 SAV sites, 2 CSH sites, 8 OSH sites, 27 GRA sites and 14 CRO sites. Wetlands were excluded from this study, and C_4 sites (Supplementary Table 1) were analysed separately (Extended Data Fig. 2). All data were constrained to the growing season using a consistent temperature threshold set at $>10^\circ\text{C}$ (5°C was also tested; Supplementary Fig. 13a), considering CRO inclusion.

Deriving community plant traits from EEO

We applied EEO theory to simulate site-specific LMA, LL and $V_{\text{cmax}25}$ (normalized to 25°C)^{26,63,80}. EEO theory hypothesizes that the relationship between LMA and LL will maximize leaf net carbon gain²⁶. Hence, the community site-mean LMA for deciduous (LMA_{de} : CRO, GRA, DBF and SAV) and evergreen (LMA_{ev} : ENF and EBF) taxa can be estimated by these equations, with detailed derivations available in previous studies^{26,68}:

$$\ln(\text{LMA}_{\text{de}}) = \ln(f) + \ln(I_{\text{abs}}) - 0.052T_a + \beta_{a,\text{de}}\alpha_{\text{bess}} + \beta_{0,\text{de}} \quad (3)$$

$$\ln(\text{LMA}_{\text{ev}}) = 0.25 \ln(f) + 0.5 \ln(I_{\text{abs}}) - 0.013T_a + \beta_{a,\text{ev}}\alpha_{\text{bess}} + \beta_{0,\text{ev}} \quad (4)$$

where f is the fraction of growing season (defined as the period with daily mean temperature $>10^\circ\text{C}$) length to a year; I_{abs} (mol photons per m^2 per day) is the mean growing season canopy APAR; T_a (K) is the mean growing season air temperature; $\beta_{a,\text{ev}}$, $\beta_{a,\text{de}}$, $\beta_{0,\text{ev}}$ and $\beta_{0,\text{de}}$ are empirical coefficients derived from Glopnet²² and China Plant Trait Database⁶⁴ (Supplementary Table 2); and α_{bess} is an ESI computed as the ratio of actual to potential evapotranspiration (ET/PET) simulated using the process model Breathing Earth System Simulator (BESS v.2.0)⁵⁴. Due to limitations in trait data, direct validation of LMA across FLUXNET2015 sites is still challenging; however, validation of EEO-derived LMA against global plant trait databases shows good performance⁶⁸.

Similarly, LL for deciduous (LL_{de}) and evergreen (LL_{ev}) taxa were estimated at the growing season scale by²⁶:

$$\text{LL}_{\text{de}} = 365f \quad (5)$$

$$\text{LL}_{\text{ev}} = \text{LMA}_{\text{ev}} \sqrt{\frac{2u C_c h_T m_c}{\varphi_0 m I_{\text{abs}} k_1 f}} \quad (6)$$

where u (768, unitless) is a constant that reflects the ageing rate, C_c (13.23 g C per g C) is a multiplier related to the carbon costs of leaf construction, k_1 (30 g biomass per mol C) is a scaling factor and h_T is the modified Arrhenius function for V_{cmax} (ref. 81):

$$h_T = \exp\left[\frac{E(T_a - 298)}{298RT_a}\right] \frac{1 + \exp\left(\frac{298\Delta S - H}{298R}\right)}{1 + \exp\left(\frac{T_a\Delta S - H}{T_a R}\right)} \quad (7)$$

where E (72 kJ mol^{-1}) and H (200 kJ mol^{-1}) are the energy of activation and deactivation of V_{cmax} , respectively⁸¹, and ΔS ($\text{kJ mol}^{-1}^\circ\text{C}^{-1}$) is an entropy factor^{63,82}:

$$\Delta S = \frac{668.39 - 1.07T_a}{1,000} \quad (8)$$

LMA and LL for other PFTs (MF, WSA, CSH and OSH) were computed as the mean value between deciduous and evergreen taxa. In equation (6), m_c is the CO_2 limitation term for the Rubisco-limited photosynthetic rate¹⁵:

$$m_c = (C_i - \Gamma^*) / (C_i + K) \quad (9)$$

where C_i (Pa) is intercellular CO_2 concentration⁸⁰:

$$C_i = C_a P_s \chi \quad (10)$$

$$\chi = \frac{\sqrt{\frac{240(K + \Gamma^*)}{1.6\eta^*}} + \frac{\Gamma^* \sqrt{\text{VPD}}}{C_a}}{\sqrt{\frac{240(K + \Gamma^*)}{1.6\eta^*}} + \sqrt{\text{VPD}}} \quad (11)$$

where P_s (Pa) is the surface pressure, C_a ($\mu\text{mol mol}^{-1}$) is the ambient CO_2 concentration⁸⁰, χ is the ratio of leaf internal to ambient CO_2 , K (Pa) is the Michaelis–Menten coefficient of Rubisco, Γ^* (Pa) is the CO_2 compensation point in the absence of dark respiration and η^* is the viscosity of water relative to its value at 25°C . K , Γ^* and η^* are functions of temperature^{80,83,84}:

$$\Gamma^* = 10^{-6} \exp\left(19.02 - \frac{37.83}{RT_a}\right) P_s \quad (12)$$

$$K = 10^{-6} \exp\left(38.05 - \frac{79.43}{RT_a}\right) \left[1 + \frac{0.21P_s}{10^{-3} \exp\left(20.30 - \frac{36.38}{RT_a}\right) P_s}\right] P_s \quad (13)$$

$$\eta^* = \exp\left[-\frac{580(T_a - 298)}{(-138 + T_a)^2}\right] \quad (14)$$

where R ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$) is the universal gas constant.

In equation (6), φ_0 is the intrinsic quantum yield of photosynthesis (g C mol^{-1})^{63,84}:

$$\varphi_0 = \text{fAPAR} \varphi_{\text{PSII,max,l}} b_l M / k \quad (15)$$

$$\varphi_{\text{PSII,max,l}} = 0.352 + 0.022(T_a - 273.16) - 0.00034(T_a - 273.16)^2 \quad (16)$$

where fAPAR denotes the dynamic leaf absorptance, $\varphi_{\text{PSII,max,l}}$ is the maximum light-adapted quantum yield of photosystem II, b_l is the ratio (0.5) between photosystem-II-captured light and leaf-absorbed light, M (12 g mol^{-1}) is the weight of one mol of carbon and k (4, unitless) is the quantity of electron equivalents needed to reduce a single molecule of CO_2 .

In equation (6), m is the CO_2 limitation term for the electron-transport-limited photosynthetic rate^{15,26}:

$$m = (C_i - \Gamma^*) / (C_i + 2\Gamma^*) \quad (17)$$

Moreover, EEO hypothesizes a synergistic optimization of Rubisco, light and water consumption for photosynthesis⁸⁰. Together with the coordination theory, which balances Rubisco- and electron-transport-limited assimilation rates⁸⁵, this will yield:

$$A_c = V_{\text{cmax}} \frac{C_i - \Gamma^*}{C_i + K} = A_j = \frac{\varphi_0 I}{\sqrt{1 + \left(\frac{4\varphi_0 I}{J_{\text{max}}}\right)^2}} \frac{C_i - \Gamma^*}{C_i + 2\Gamma^*} \quad (18)$$

where A_c and A_j are the Rubisco-limited CO_2 photosynthetic rate and the electron-transport-limited CO_2 photosynthetic rate, respectively. I is the absorbed PPFD.

Solving equation (18) and normalizing it with the Arrhenius function, the community site-mean $V_{\text{cmax}25}$ for C_3 species can be estimated as^{63,80}:

$$V_{\text{cmax}25} = \varphi_0 I^{\frac{C_i + K}{C_i + 2\Gamma^*}} \sqrt{1 - \left[\frac{4c(C_i + 2\Gamma^*)}{C_i - \Gamma^*} \right]^{\frac{2}{3}}} \frac{1}{h_T} \quad (19)$$

where c (0.103, unitless) is a cost factor estimated with typical values of $J_{\text{max}}/V_{\text{cmax}} = 1.88$ (ref. 82) and $C_i/C_a = 0.8$ (ref. 86). We incorporated a dynamic rolling window spanning from the beginning of the growing season to the target time, enabling seasonal dynamics in $V_{\text{cmax}25}$ and potentially capturing lag effects and trait acclimation⁶³. Accounting for antecedent environment conditions and dynamic light absorption has provided reasonably accurate estimation of seasonal $V_{\text{cmax}25}$ across ten diverse ecosystems⁶³. The EEO-based $V_{\text{cmax}25}$ has been incorporated into a prominent process-based model to estimate GPP, with through validation against FLUXNET data⁵⁴.

Remotely sensed community site-mean Chl

We used a recently released global eight-day Chl product at 500-m resolution covering the period from 2000 to 2020⁶². The Chl data were produced from MODIS surface reflectance, where the effects of LAI and canopy non-photosynthetic material were excluded using radiative transfer models. The Chl data have shown good agreement with ground measurements and reasonable seasonal variations in different PFTs⁶². To align the Chl data with the footprints of FLUXNET2015 sites, we extracted nine pixels, including eight adjacent to the site pixel, and computed their mean as the community site value ($1.5 \times 1.5 \text{ km}^2$). The eight-day Chl data were further interpolated to a daily scale using the Akima method for gaps smaller than 50%. Moreover, to assess uncertainties stemming from the choice of Chl products, we compared the results with an additional seven-day MERIS Chl product at 300-m resolution spanning from 2003 to 2012 (ref. 61) (Extended Data Fig. 4 and Supplementary Fig. 10). We processed the MERIS Chl data to a daily scale using the Akima method⁷⁷ and resampled them to a 900-m resolution.

Canopy structure

LAI data were obtained from MODIS MOD15A2H (eight-day, 18 February 2000 to 3 July 2002) and MCD15A3H (four-day, 4 July 2002 onwards) v.6.1 products, both at a $1.0 \times 1.0 \text{ km}^2$ resolution. The original data were quality controlled (FparLai_QC and FparExtra_QC) and smoothed with a Savitzky–Golay filter. The LAI data were further interpolated to a daily scale using the Akima interpolator for gaps smaller than 50%⁷⁷. Similarly, Cl_{clump} data at the site level were obtained from a global 500-m product derived from MODIS BRDF data⁸⁷. We applied the same method as MODIS LAI to derive daily Cl_{clump} . Moreover, to assess uncertainties stemming from the choice of LAI products, we compared the results with additional GLASS LAI products at a daily and $1.0 \times 1.0 \text{ km}^2$ resolution⁸⁸ (Supplementary Fig. 11).

Developing and screening machine learning models

We deployed three candidate machine learning models, including deep neural networks, light gradient boosting machine (LightGBM) and XGBoost, to associate leaf traits (Chl, LL, $V_{\text{cmax}25}$ and LMA), canopy structure (LAI and Cl_{clump}) and meteorological variables (Cl_{cloud} , T_a , SWC, VPD, ESI and CO_2) with ecosystem PE. CO_2 was included to assess potential fertilization effects. The complete information for all explanatory variables is provided in Supplementary Section 1.2. To prevent collinearity and redundancy among input variables, we performed a variance inflation factor (VIF) analysis, setting a threshold of $\text{VIF} < 10$ (Supplementary Table 3), and a non-parametric Spearman correlation analysis with a threshold of $\rho < 0.75$ (Supplementary

Fig. 2d). Detailed information about the structure and parameters for each model is shown in our publicly available code. Starting with the default hyperparameter values (for example, for XGBoost, $n_{\text{estimators}}$, 100; max_depth , 6; and learning_rate , 0.3), we tuned one parameter at a time until further adjustments no longer resulted in notable improvements in model performance. We developed the deep neural network model using a systematic approach, starting with a shallow network of three hidden layers and gradually increasing the depth on the basis of performance. Neuron counts were selected as follows: 128, 128, 64, 64, 32, 32, 16, 16, 8 and 8 to balance model complexity and computational efficiency, with reduced neuron counts in deeper layers to minimize overfitting. ReLU activation was used in the hidden layers, as it outperformed the Tanh function, effectively addressing the vanishing gradient problem. To ensure reliable estimates of model performance, we implemented a k -fold cross-validation scheme with k set to either five or ten. In each iteration, 80% (for $k = 5$) or 90% (for $k = 10$) of the data were used for training, while the remaining 20% or 10% were used for validation. The evaluation matrix includes R^2 , RMSE, the slope of the regression line and the bias of estimation. On the basis of the final averaged matrix (Supplementary Fig. 5), we selected XGBoost to further investigate the underlying mechanisms governing ecosystem-scale PE ('Mechanistic interpretation of machine learning models'). Although LightGBM produced comparable performance, XGBoost was preferred due to its robust handling of small to medium-sized datasets, finer control over regularization and proven success in similar ecological studies^{42,89,90}.

Mechanistic interpretation of machine learning models

To quantify and interpret the contributions, potential nonlinearity and interactions among leaf traits, canopy structure and climate on ecosystem PE, we integrated cooperative game theory principles (SHAP values) into the black-box machine learning models. Cooperative game theory deals with situations where groups of players cooperate to achieve a common goal, often by forming coalitions⁹¹. In this context, each feature (for example, Chl) in a machine learning model can be seen as a player, and the goal is to cooperate with other variables to estimate ecosystem PE accurately. Here we used the SHAP TreeExplainer to compute SHAP values for decision-tree-based models (XGBoost and LightGBM) and KernelExplainer for the deep neural network model^{91,92}. SHAP is a state-of-the-art model-agnostic method that not only offers local explanations but also interprets the global model structure on the basis of numerous local explanations. SHAP values provide a fair attribution of importance by considering all possible combinations of input variables, ensuring that each trait's influence is evaluated in the context of others. This approach therefore enables us to unveil the primary effects (SHAP values) of each feature as well as potential interactive effects (SHAP interaction values) in a theoretically grounded way⁹¹. XGBoost and SHAP methods have been successfully applied in multidisciplinary studies^{42,89,90}.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets used in this study are available at <https://fluxnet.org/data/fluxnet2015-dataset/> (FLUXNET2015), <https://modis.gsfc.nasa.gov/data/dataproduct/mod15.php> (MODIS LAI/fPAR), <https://land.copernicus.eu/en/products/vegetation> (GEOV2 fPAR), <http://www.glass.umd.edu/Download.html> (GLASS LAI), www.geodata.cn (clumping index), <https://doi.org/10.1038/nature02403> (Gloptnet dataset) and <https://doi.org/10.1002/ecy.2091> (the China Plant Trait Database) and on Zenodo at <https://doi.org/10.5281/zenodo.5805575> (ref. 93) (MODIS chlorophyll content) and <https://doi.org/10.5281/zenodo.10467919> (ref. 94) (MERIS chlorophyll content). The data supporting Figs. 1–4 are available on Zenodo at <https://doi.org/10.5281/zenodo.14191212> (ref. 95).

Code availability

All relevant code is available on GitHub at <https://github.com/Heeheedaa/PE>.

References

- Franklin, J., Serra-Diaz, J. M., Syphard, A. D. & Regan, H. M. Global change and terrestrial plant community dynamics. *Proc. Natl Acad. Sci. USA* **113**, 3725–3734 (2016).
- Stirbet, A., Lazár, D., Guo, Y. & Govindjee, G. Photosynthesis: basics, history and modelling. *Ann. Bot.* **126**, 511–537 (2020).
- Monteith, J. L. Solar radiation and productivity in tropical ecosystems. *J. Appl. Ecol.* **9**, 747–766 (1972).
- Reichstein, M., Bahn, M., Mahecha, M. D., Kattge, J. & Baldocchi, D. D. Linking plant and ecosystem functional biogeography. *Proc. Natl Acad. Sci. USA* **111**, 13697–13702 (2014).
- Migliavacca, M. et al. The three major axes of terrestrial ecosystem function. *Nature* **598**, 468–472 (2021).
- Bonan, G. B. & Doney, S. C. Climate, ecosystems, and planetary futures: the challenge to predict life in Earth system models. *Science* **359**, eaam8328 (2018).
- Steffen, W. et al. The emergence and evolution of Earth system science. *Nat. Rev. Earth Environ.* **1**, 54–63 (2020).
- Chen, J.-H. et al. Nuclear-encoded synthesis of the D1 subunit of photosystem II increases photosynthetic efficiency and crop yield. *Nat. Plants* **6**, 570–580 (2020).
- Bloomfield, K. J., Stocker, B. D., Keenan, T. F. & Prentice, I. C. Environmental controls on the light use efficiency of terrestrial gross primary production. *Glob. Change Biol.* **29**, 1037–1053 (2023).
- He, M. et al. Global spectrum of vegetation light-use efficiency. *Geophys. Res. Lett.* **49**, e2022GL099550 (2022).
- Stocker, B. D. et al. Quantifying soil moisture impacts on light use efficiency across biomes. *New Phytol.* **218**, 1430–1449 (2018).
- Gu, L. et al. Advantages of diffuse radiation for terrestrial ecosystem productivity. *J. Geophys. Res. Atmos.* **107**, ACL 2-1–ACL 2-23 (2002).
- Balzarolo, M. et al. Different determinants of radiation use efficiency in cold and temperate forests. *Glob. Ecol. Biogeogr.* **28**, 1649–1667 (2019).
- Berry, J. A. & Downton, W. J. S. in *Photosynthesis V2: Development, Carbon Metabolism, and Plant Productivity* (ed. Govindjee) Ch. 9 (Academic Press, 1982).
- Farquhar, G. D., von Caemmerer, S. & Berry, J. A. A biochemical model of photosynthetic CO₂ assimilation in leaves of C₃ species. *Planta* **149**, 78–90 (1980).
- Hikosaka, K., Ishikawa, K., Borjigidai, A., Muller, O. & Onoda, Y. Temperature acclimation of photosynthesis: mechanisms involved in the changes in temperature dependence of photosynthetic rate. *J. Exp. Bot.* **57**, 291–302 (2006).
- Scafaro, A. P., Posch, B. C., Evans, J. R., Farquhar, G. D. & Atkin, O. K. Rubisco deactivation and chloroplast electron transport rates co-limit photosynthesis above optimal leaf temperature in terrestrial plants. *Nat. Commun.* **14**, 2820 (2023).
- Pinheiro, C. & Chaves, M. M. Photosynthesis and drought: can we make metabolic connections from available data? *J. Exp. Bot.* **62**, 869–882 (2011).
- Chaves, M. M. et al. How plants cope with water stress in the field? Photosynthesis and growth. *Ann. Bot.* **89**, 907–916 (2002).
- Tezara, W., Mitchell, V. J., Driscoll, S. D. & Lawlor, D. W. Water stress inhibits plant photosynthesis by decreasing coupling factor and ATP. *Nature* **401**, 914–917 (1999).
- Feng, X. & Dietze, M. Scale dependence in the effects of leaf ecophysiological traits on photosynthesis: Bayesian parameterization of photosynthesis models. *New Phytol.* **200**, 1132–1144 (2013).
- Wright, I. J. et al. The worldwide leaf economics spectrum. *Nature* **428**, 821–827 (2004).
- Osnas, J. L. D., Lichstein, J. W., Reich, P. B. & Pacala, S. W. Global leaf trait relationships: mass, area, and the leaf economics spectrum. *Science* **340**, 741–744 (2013).
- Funk, J. L. et al. Revisiting the Holy Grail: using plant functional traits to understand ecological processes. *Biol. Rev.* **92**, 1156–1173 (2017).
- Eberhard, S., Finazzi, G. & Wollman, F.-A. The dynamics of photosynthesis. *Annu. Rev. Genet.* **42**, 463–515 (2008).
- Wang, H. et al. Leaf economics fundamentals explained by optimality principles. *Sci. Adv.* **9**, eadd5667 (2023).
- Kikuzawa, K., Onoda, Y., Wright, I. J. & Reich, P. B. Mechanisms underlying global temperature-related patterns in leaf longevity. *Glob. Ecol. Biogeogr.* **22**, 982–993 (2013).
- Marqués, L. et al. Acclimation of phenology relieves leaf longevity constraints in deciduous forests. *Nat. Ecol. Evol.* **7**, 198–204 (2023).
- Kikuzawa, K. & Lechowicz, M. J. Toward synthesis of relationships among leaf longevity, instantaneous photosynthetic rate, lifetime leaf carbon gain, and the gross primary production of forests. *Am. Nat.* **168**, 373–383 (2006).
- Li, F. et al. Vegetation clumping modulates global photosynthesis through adjusting canopy light environment. *Glob. Change Biol.* **29**, 731–746 (2023).
- Musavi, T. et al. Potential and limitations of inferring ecosystem photosynthetic capacity from leaf functional traits. *Ecol. Evol.* **6**, 7352–7366 (2016).
- Dechant, B. et al. Intercomparison of global foliar trait maps reveals fundamental differences and limitations of upscaling approaches. *Remote Sens. Environ.* **311**, 114276 (2024).
- Yan, P., He, N., Yu, K., Xu, L. & Van Meerbeek, K. Integrating multiple plant functional traits to predict ecosystem productivity. *Commun. Biol.* **6**, 239 (2023).
- Wang, Y. P. et al. Correlations among leaf traits provide a significant constraint on the estimate of global gross primary production. *Geophys. Res. Lett.* **39** (2012).
- Fyllas, N. M. et al. Solar radiation and functional traits explain the decline of forest primary productivity along a tropical elevation gradient. *Ecol. Lett.* **20**, 730–740 (2017).
- van der Plas, F. et al. Plant traits alone are poor predictors of ecosystem properties and long-term ecosystem functioning. *Nat. Ecol. Evol.* **4**, 1602–1611 (2020).
- Hagan, J. G., Henn, J. J. & Osterman, W. H. A. Plant traits alone are good predictors of ecosystem properties when used carefully. *Nat. Ecol. Evol.* **7**, 332–334 (2023).
- van der Plas, F. et al. Reply to: Plant traits alone are good predictors of ecosystem properties when used carefully. *Nat. Ecol. Evol.* **7**, 335–336 (2023).
- Garbulsky, M. F. et al. Patterns and controls of the variability of radiation use efficiency and primary productivity across terrestrial ecosystems. *Glob. Ecol. Biogeogr.* **19**, 253–267 (2010).
- Chasmer, L. et al. Investigating light-use efficiency across a jack pine chronosequence during dry and wet years. *Tree Physiol.* **28**, 1395–1406 (2008).
- Tang, S. et al. Global patterns and climate controls of terrestrial ecosystem light use efficiency. *J. Geophys. Res. Biogeosci.* **125**, e2020JG005908 (2020).
- Wang, H. et al. Exploring complex water stress–gross primary production relationships: impact of climatic drivers, main effects, and interactive effects. *Glob. Change Biol.* **28**, 4110–4123 (2022).
- Liu, L. et al. Soil moisture dominates dryness stress on ecosystem production globally. *Nat. Commun.* **11**, 4892 (2020).
- Long, S. P. et al. Into the shadows and back into sunlight: photosynthesis in fluctuating light. *Annu. Rev. Plant Biol.* **73**, 617–648 (2022).

45. Rap, A. et al. Fires increase Amazon forest productivity through increases in diffuse radiation. *Geophys. Res. Lett.* **42**, 4654–4662 (2015).
46. Rap, A. et al. Enhanced global primary production by biogenic aerosol via diffuse radiation fertilization. *Nat. Geosci.* **11**, 640–644 (2018).
47. Zhou, H. et al. Large contributions of diffuse radiation to global gross primary productivity during 1981–2015. *Glob. Biogeochem. Cycles* **35**, e2021GB006957 (2021).
48. Fan, Y. et al. Solar geoengineering can alleviate climate change pressures on crop yields. *Nat. Food* **2**, 373–381 (2021).
49. Dechant, B. et al. Canopy structure explains the relationship between photosynthesis and sun-induced chlorophyll fluorescence in crops. *Remote Sens. Environ.* **241**, 111733 (2020).
50. Migliavacca, M. et al. Plant functional traits and canopy structure control the relationship between photosynthetic CO₂ uptake and far-red sun-induced fluorescence in a Mediterranean grassland under different nutrient availability. *New Phytol.* **214**, 1078–1091 (2017).
51. Niinemets, U. Photosynthesis and resource distribution through plant canopies. *Plant Cell Environ.* **30**, 1052–1071 (2007).
52. Zhuang, J., Zhou, L., Wang, Y. & Chi, Y. Nitrogen allocation regulates the relationship between maximum carboxylation rate and chlorophyll content along the vertical gradient of subtropical forest canopy. *Agric. For. Meteorol.* **307**, 108512 (2021).
53. Li, Y., Song, H., Zhou, L., Xu, Z. & Zhou, G. Vertical distributions of chlorophyll and nitrogen and their associations with photosynthesis under drought and rewatering regimes in a maize field. *Agric. For. Meteorol.* **272–273**, 40–54 (2019).
54. Li, B. et al. BESSv2.0: a satellite-based and coupled-process model for quantifying long-term global land–atmosphere fluxes. *Remote Sens. Environ.* **295**, 113696 (2023).
55. Leng, J. et al. Global datasets of hourly carbon and water fluxes simulated using a satellite-based process model with dynamic parameterizations. *Earth Syst. Sci. Data* **16**, 1283–1300 (2024).
56. Luo, X., Croft, H., Chen, J. M., He, L. & Keenan, T. F. Improved estimates of global terrestrial photosynthesis using information on leaf chlorophyll content. *Glob. Change Biol.* **25**, 2499–2514 (2019).
57. Peaucelle, M. et al. Covariations between plant functional traits emerge from constraining parameterization of a terrestrial biosphere model. *Glob. Ecol. Biogeogr.* **28**, 1351–1365 (2019).
58. Poorter, H., Niinemets, Ü., Poorter, L., Wright, I. J. & Villar, R. Causes and consequences of variation in leaf mass per area (LMA): a meta-analysis. *New Phytol.* **182**, 565–588 (2009).
59. Walker, A. P. et al. The relationship of leaf photosynthetic traits— V_{cmax} and J_{max} —to leaf nitrogen, leaf phosphorus, and specific leaf area: a meta-analysis and modeling study. *Ecol. Evol.* **4**, 3218–3235 (2014).
60. Berzaghi, F. et al. Towards a new generation of trait-flexible vegetation models. *Trends Ecol. Evol.* **35**, 191–205 (2020).
61. Xu, M. et al. Retrieving global leaf chlorophyll content from MERIS data using a neural network method. *ISPRS J. Photogramm. Remote Sens.* **192**, 66–82 (2022).
62. Xu, M. et al. A 21-year time series of global leaf chlorophyll content maps from MODIS imagery. *IEEE Trans. Geosci. Remote Sens.* **60**, 1–13 (2022).
63. Jiang, C., Ryu, Y., Wang, H. & Keenan, T. F. An optimality-based model explains seasonal variation in C₃ plant photosynthetic capacity. *Glob. Change Biol.* **26**, 6493–6510 (2020).
64. Wang, H. et al. The China Plant Trait Database: toward a comprehensive regional compilation of functional traits for land plants. *Ecology* **99**, 500 (2018).
65. Tian, D. et al. Global leaf nitrogen and phosphorus stoichiometry and their scaling exponent. *Natl Sci. Rev.* **5**, 728–739 (2018).
66. Crous, K. Y. et al. Nitrogen and phosphorus availabilities interact to modulate leaf trait scaling relationships across six plant functional types in a controlled-environment study. *New Phytol.* **215**, 992–1008 (2017).
67. Ellsworth, D. S. et al. Convergence in phosphorus constraints to photosynthesis in forests around the world. *Nat. Commun.* **13**, 5005 (2022).
68. Dong, N., Dechant, B., Wang, H., Wright, I. J. & Prentice, I. C. Global leaf-trait mapping based on optimality theory. *Glob. Ecol. Biogeogr.* **32**, 1152–1162 (2023).
69. Luo, X. et al. Global variation in the fraction of leaf nitrogen allocated to photosynthesis. *Nat. Commun.* **12**, 4866 (2021).
70. Zhang, Y., Migliavacca, M., Penuelas, J. & Ju, W. Advances in hyperspectral remote sensing of vegetation traits and functions. *Remote Sens. Environ.* **252**, 112121 (2021).
71. Gu, L., Han, J., Wood, J. D., Chang, C. Y. Y. & Sun, Y. Sun-induced Chl fluorescence and its importance for biophysical modeling of photosynthesis based on light reactions. *New Phytol.* **223**, 1179–1191 (2019).
72. Billing, M., Sakschewski, B., von Bloh, W., Vogel, J. & Thonicke, K. ‘How to adapt forests?’ Exploring the role of leaf trait diversity for long-term forest biomass under new climate normals. *Glob. Change Biol.* **30**, e17258 (2024).
73. Prentice, I. C. et al. Principles for satellite monitoring of vegetation carbon uptake. *Nat. Rev. Earth Environ.* **5**, 818–832 (2024).
74. Reichstein, M. et al. Deep learning and process understanding for data-driven Earth system science. *Nature* **566**, 195–204 (2019).
75. Pastorello, G. et al. The FLUXNET2015 dataset and the ONEFlux processing pipeline for eddy covariance data. *Sci. Data* **7**, 225 (2020).
76. Reichstein, M. et al. On the separation of net ecosystem exchange into assimilation and ecosystem respiration: review and improved algorithm. *Glob. Change Biol.* **11**, 1424–1439 (2005).
77. Akima, H. A new method of interpolation and smooth curve fitting based on local procedures. *J. ACM* **17**, 589–602 (1970).
78. Verger, A. et al. GEOV2: improved smoothed and gap filled time series of LAI, FAPAR and FCOVER 1km Copernicus Global Land products. *Int. J. Appl. Earth Obs. Geoinf.* **123**, 103479 (2023).
79. Meek, D. W., Hatfield, J. L., Howell, T. A., Idso, S. B. & Reginato, R. J. A generalized relationship between photosynthetically active radiation and solar radiation. *Agron. J.* **76**, 939–945 (1984).
80. Wang, H. et al. Towards a universal model for carbon dioxide uptake by plants. *Nat. Plants* **3**, 734–741 (2017).
81. Medlyn, B. E. et al. Temperature response of parameters of a biochemically based model of photosynthesis. II. A review of experimental data. *Plant Cell Environ.* **25**, 1167–1179 (2002).
82. Kattge, J. & Knorr, W. Temperature acclimation in a biochemical model of photosynthesis: a reanalysis of data from 36 species. *Plant Cell Environ.* **30**, 1176–1190 (2007).
83. Bernacchi, C. J., Singsaas, E. L., Pimentel, C., Portis, A. R. Jr & Long, S. P. Improved temperature response functions for models of Rubisco-limited photosynthesis. *Plant Cell Environ.* **24**, 253–259 (2001).
84. Bernacchi, C. J., Pimentel, C. & Long, S. P. In vivo temperature response functions of parameters required to model RuBP-limited photosynthesis. *Plant Cell Environ.* **26**, 1419–1430 (2003).
85. Chen, J.-L., Reynolds, J. F., Harley, P. C. & Tenhunen, J. D. Coordination theory of leaf nitrogen distribution in a canopy. *Oecologia* **93**, 63–69 (1993).
86. Lloyd, J. & Farquhar, G. D. ¹³C discrimination during CO₂ assimilation by the terrestrial biosphere. *Oecologia* **99**, 201–215 (1994).
87. Wei, S., Fang, H., Schaaf, C. B., He, L. & Chen, J. M. Global 500m clumping index product derived from MODIS BRDF data (2001–2017). *Remote Sens. Environ.* **232**, 111296 (2019).

88. Ma, H. & Liang, S. Development of the GLASS 250-m leaf area index product (version 6) from MODIS data using the bidirectional LSTM deep learning model. *Remote Sens. Environ.* **273**, 112985 (2022).
89. Lai, Y. et al. Global change progressively increases foliar nitrogen–phosphorus ratios in China's subtropical forests. *Glob. Change Biol.* **30**, e17201 (2024).
90. Yan, Y. et al. Climate-induced tree-mortality pulses are obscured by broad-scale and long-term greening. *Nat. Ecol. Evol.* **8**, 912–923 (2024).
91. Lundberg, S. M. et al. From local explanations to global understanding with explainable AI for trees. *Nat. Mach. Intell.* **2**, 56–67 (2020).
92. Lundberg, S. M. & Lee, S.-I. A unified approach to interpreting model predictions. *Adv. Neural Inf. Process. Syst.* <https://doi.org/10.48550/arXiv.1705.07874> (2017).
93. Xu, M., Liu, R., Chen, J. M., Liu, Y. & Shang, R. Global leaf chlorophyll content (LCC) product from MODIS imagery (2000–2020). *Zenodo* <https://doi.org/10.5281/zenodo.5805575> (2021).
94. Xu, M., Liu, R., Chen, J. M., Shang, R. & Liu, Y. Global leaf chlorophyll content product from MERIS imagery (GLOBMAP MERIS LCC). *Zenodo* <https://doi.org/10.5281/zenodo.10467919> (2022).
95. Yan, Y. Plant traits shape global spatiotemporal variations in photosynthetic efficiency. *Zenodo* <https://doi.org/10.5281/zenodo.14191212> (2024).

Acknowledgements

We appreciate the valuable suggestions from C. Chen and F. Ma at Seoul National University during the preparation of this paper and wish them success in securing a postdoctoral position. This work was supported by the China Postdoctoral Science Foundation (grant no. 2023M740631), the National Key Research and Development Program of the Ministry of Science and Technology (grant no. 2020YFA0608701), the National Natural Science Foundation of China (grant nos. 42471356, 42171353, U23A2002 and 42201360), and sDiv, the Synthesis Centre of the German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig (DFG FZT 118, 202548816).

Author contributions

YY., B.L., B.D. and J.M.C. conceived and designed the study. YY. performed the analysis and drafted the paper, with input from all co-authors. B.L. prepared the clumping index and the MODIS LAI data and ran the BESS v.2.0 model. M.X. processed the GLASS LAI and chlorophyll data. J.L. prepared the GEOV2 fPAR data. All authors contributed to discussions and revisions of the paper.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41477-025-01958-2>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41477-025-01958-2>.

Correspondence and requests for materials should be addressed to Yulin Yan or Jing M. Chen.

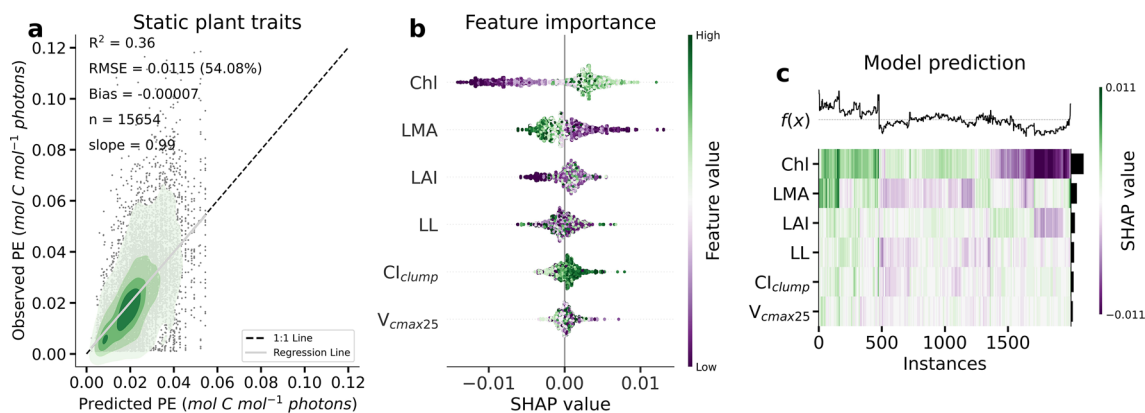
Peer review information *Nature Plants* thanks Tyler Davis, Christiane Werner and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

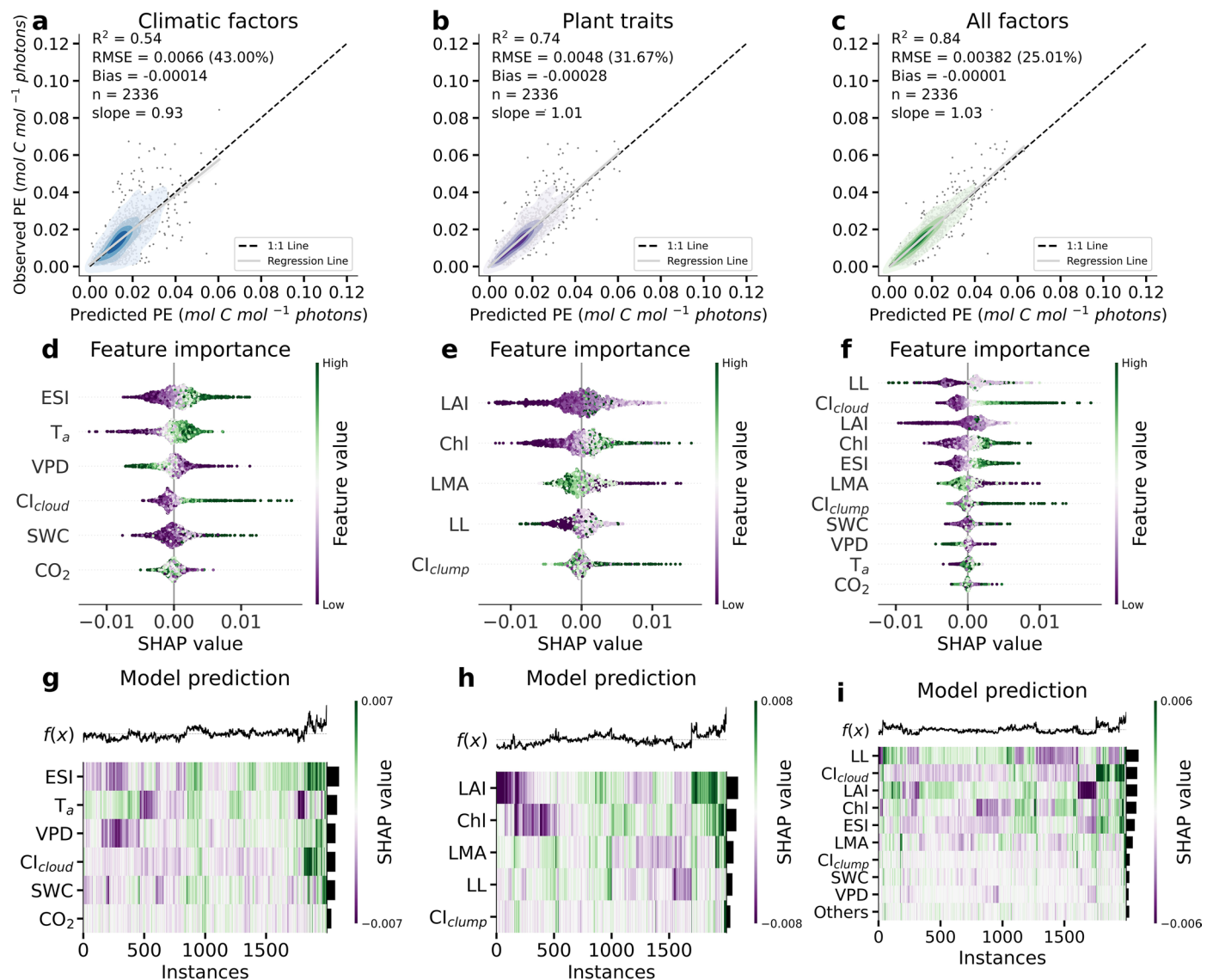
Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2025



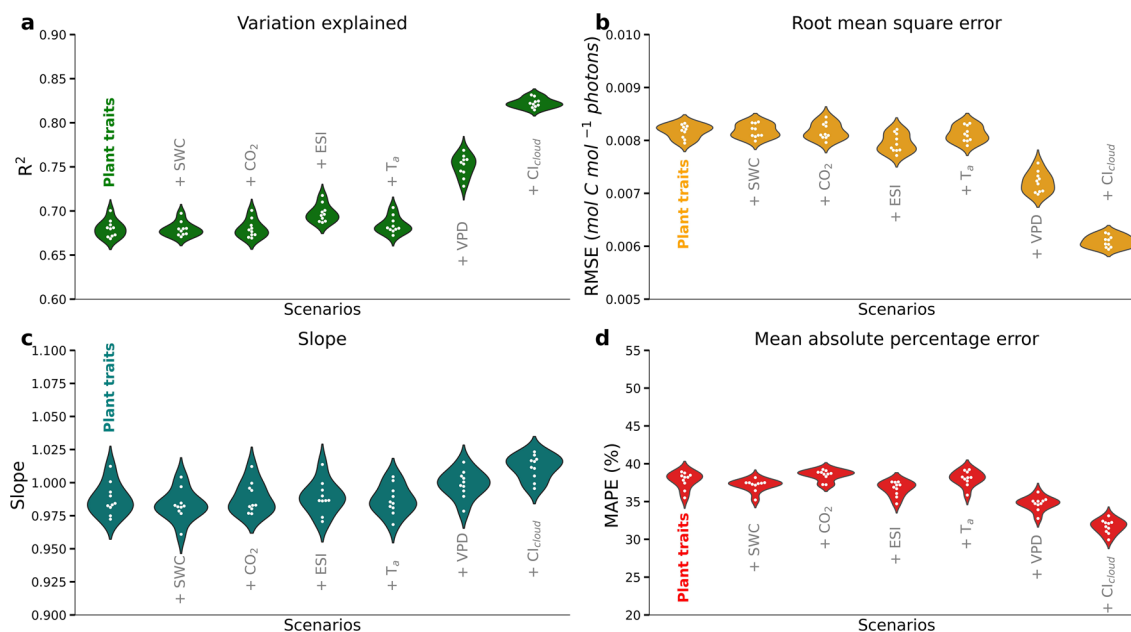
Extended Data Fig. 1 | Explanatory power of static leaf traits (representing the growing season mean) only in estimating daily ecosystem-scale PE for C_3 vegetation. (a) Shaded area indicates Gaussian kernel density estimation. **(b)** Contribution (SHAP value) of each feature ranked in descending order of importance, along with their relative feature values. **(c)** Heatmap showing clustering of 2,000 random instances by their local explanation embedding.

Columns represent instances, and rows represent features' SHAP values for each explanatory variable, where the $f(x)$ denotes the model predicted PE for each instance. Chl: leaf chlorophyll content; Cl_{clump} : clumping index; LAI: leaf area index; LL: leaf longevity; LMA: leaf mass per area; V_{cmax25} : maximum carboxylation rate at 25 °C.



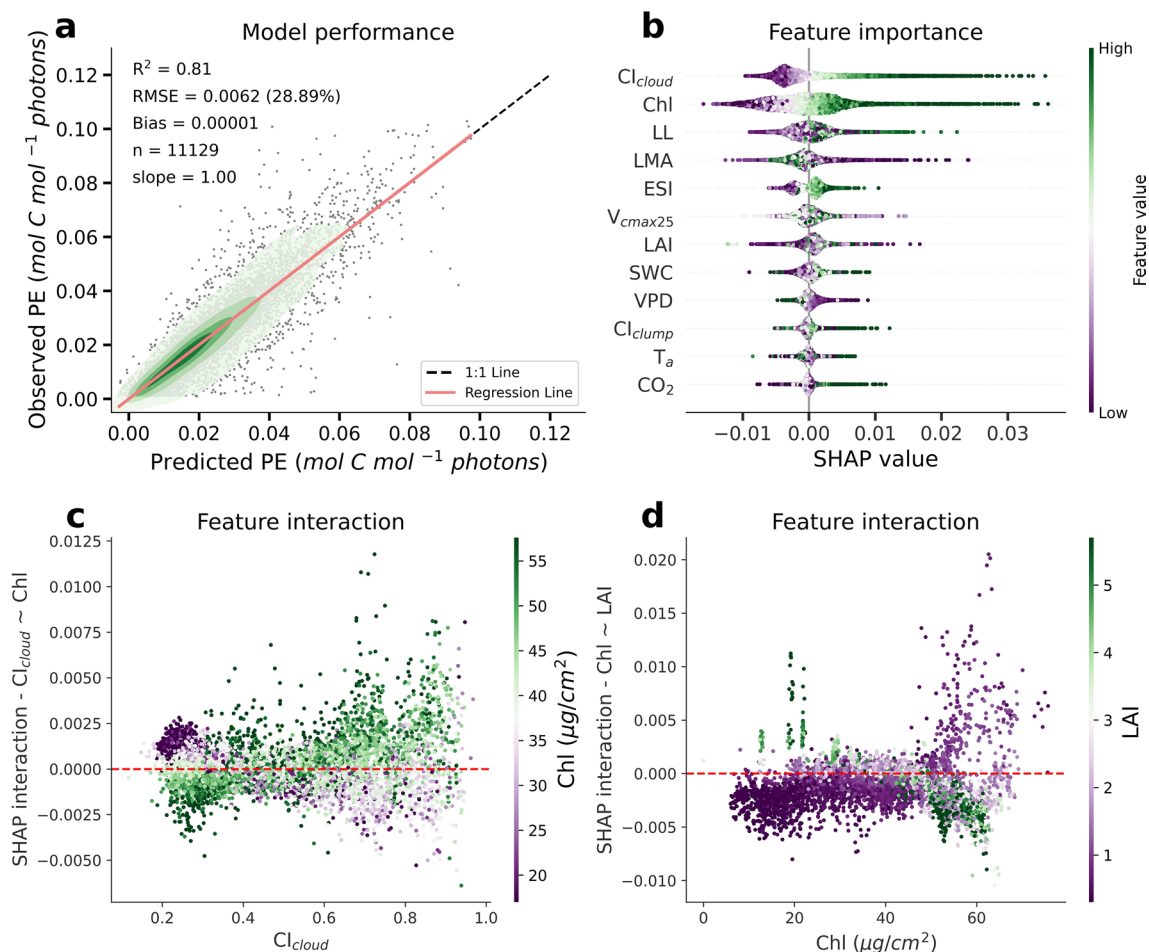
Extended Data Fig. 2 | Explanatory power of various factors in estimating daily variation in ecosystem-scale PE for C_4 vegetation. (a, d, g) climatic factors only, (b, e, h) dynamic leaf traits and canopy structure, and (c, f, i) dynamic leaf traits, climatic factors, and canopy structure; shaded area indicates Gaussian kernel density estimation. (d, e, f) Contribution (SHAP value) of each feature ranked in descending order of importance, along with their relative feature values. (g, h, i) Heatmap showing clustering of 2,000 random instances by their

local explanation embedding. Columns represent instances, and rows represent features' SHAP values for each explanatory variable, where the $f(x)$ denotes the model predicted PE for each instance. ESI: evaporative stress index; Chl: leaf chlorophyll content; Cl_{cloud} : cloudiness index; Cl_{clump} : clumping index; CO_2 : carbon dioxide; LAI: leaf area index; LL: leaf longevity; LMA: leaf mass per area; SWC: soil water content; T_a : air temperature; VPD: vapor pressure deficit.



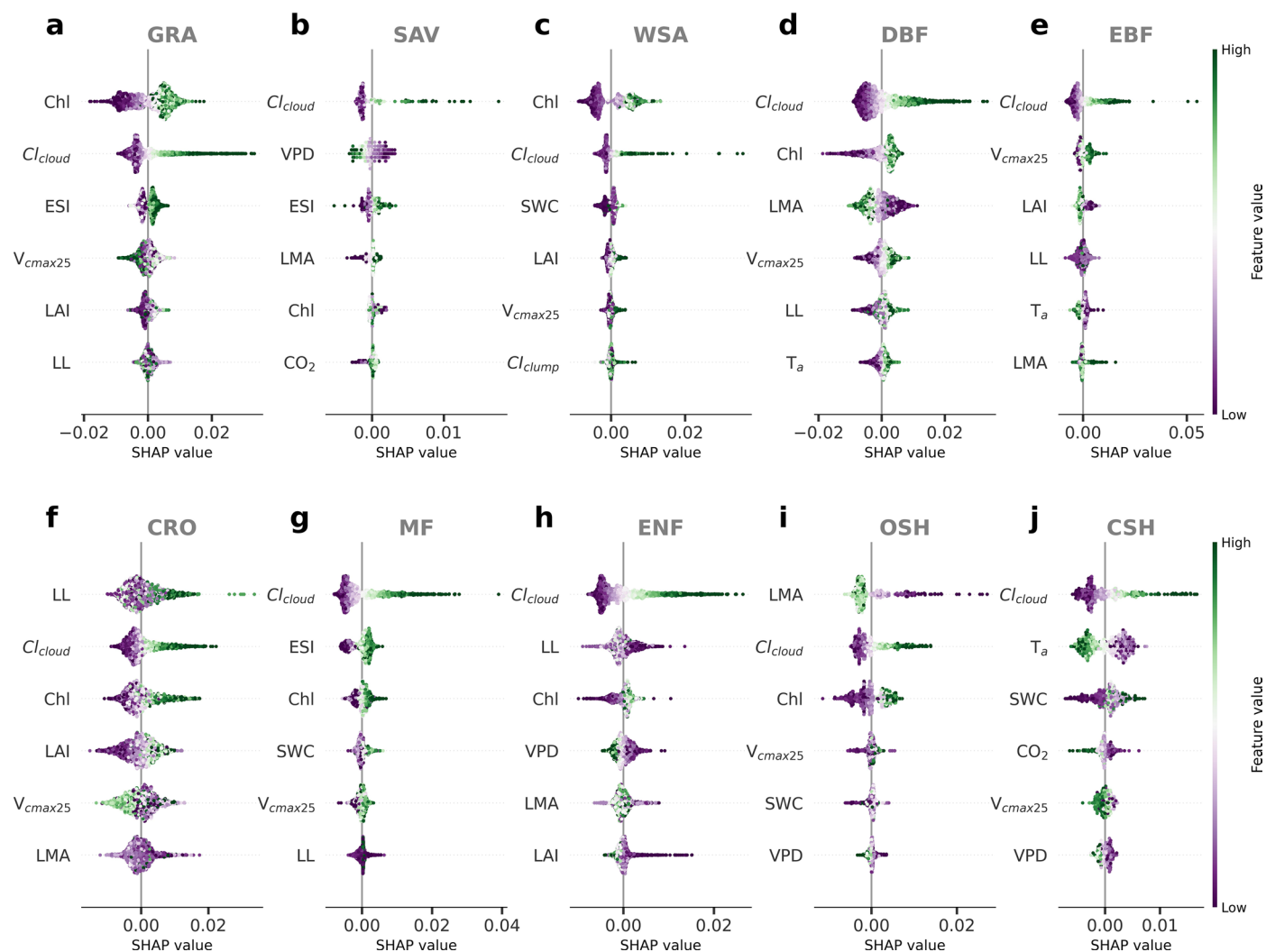
Extended Data Fig. 3 | Performance in estimating daily ecosystem-scale PE with additional contributions of climatic variables incorporated into the baseline of plant traits only model (Chl, Cl_{clump}, LAI, LL, LMA, V_{max25}) for C₃ vegetation. (a) Variation explained. (b) Root mean square error. (c) Slope of the regression lines against FLUXNET2015 observed PE. (d) Mean absolute

percentage error. ESI: evaporative stress index; Chl: leaf chlorophyll content; Cl_{cloud}: cloudiness index; Cl_{clump}: clumping index; CO₂: carbon dioxide; LAI: leaf area index; LL: leaf longevity; LMA: leaf mass per area; SWC: soil water content; T_a: air temperature; VPD: vapor pressure deficit.



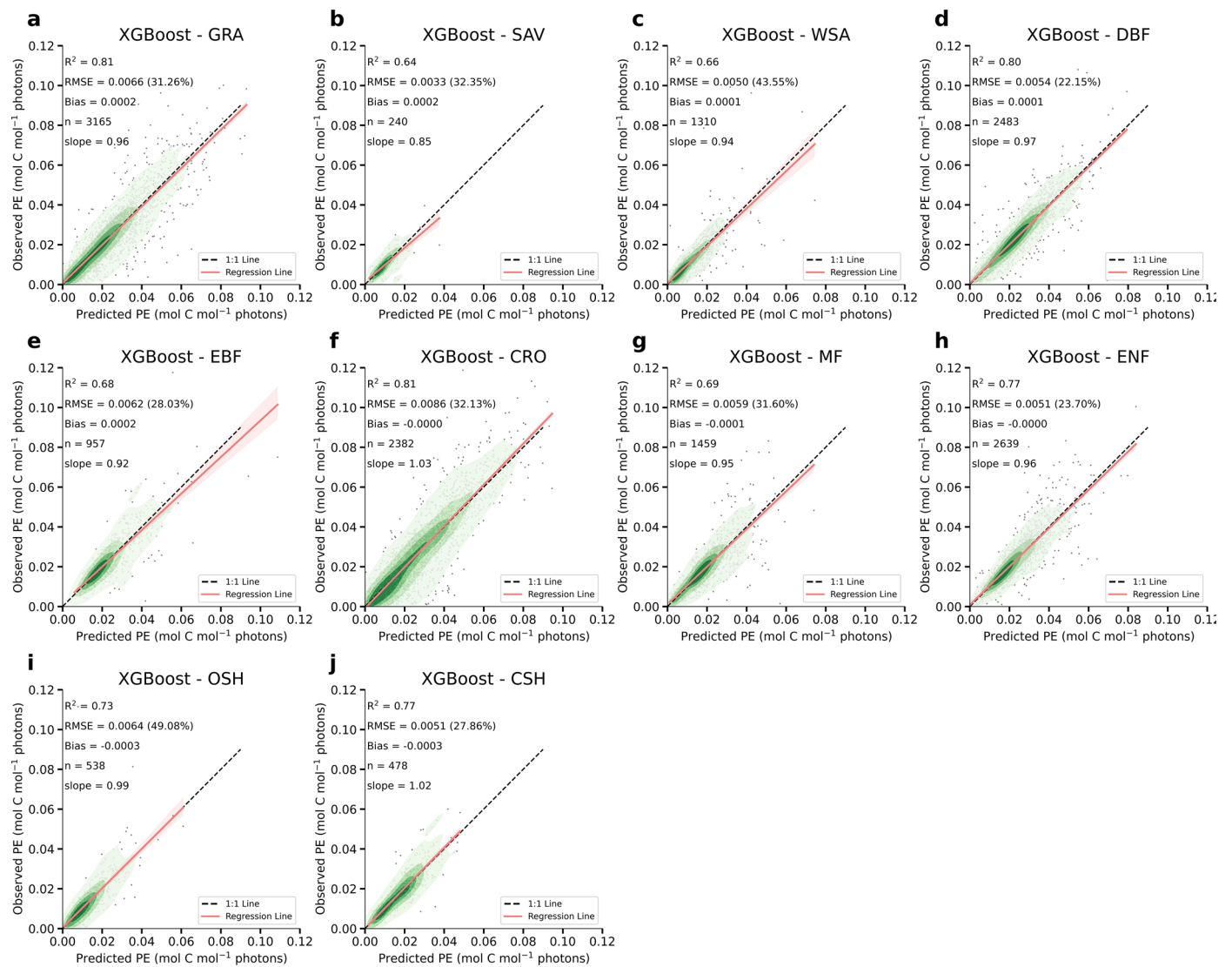
Extended Data Fig. 4 | Performance of the XGBoost model in estimating daily ecosystem-scale PE by replacing MODIS Chl product with MERIS Chl data for C_3 vegetation. (a) Model performance, (b) feature importance, (c) interaction between Cl_{cloud} and Chl content, (d) interaction between Chl content and LAI.

ESI: evaporative stress index; Chl: leaf chlorophyll content; Cl_{cloud} : cloudiness index; Cl_{clump} : clumping index; CO_2 : carbon dioxide; LAI: leaf area index; LL: leaf longevity; LMA: leaf mass per area; SWC: soil water content; T_a : air temperature; V_{cmax25} : maximum carboxylation rate at 25 °C; VPD: vapor pressure deficit.



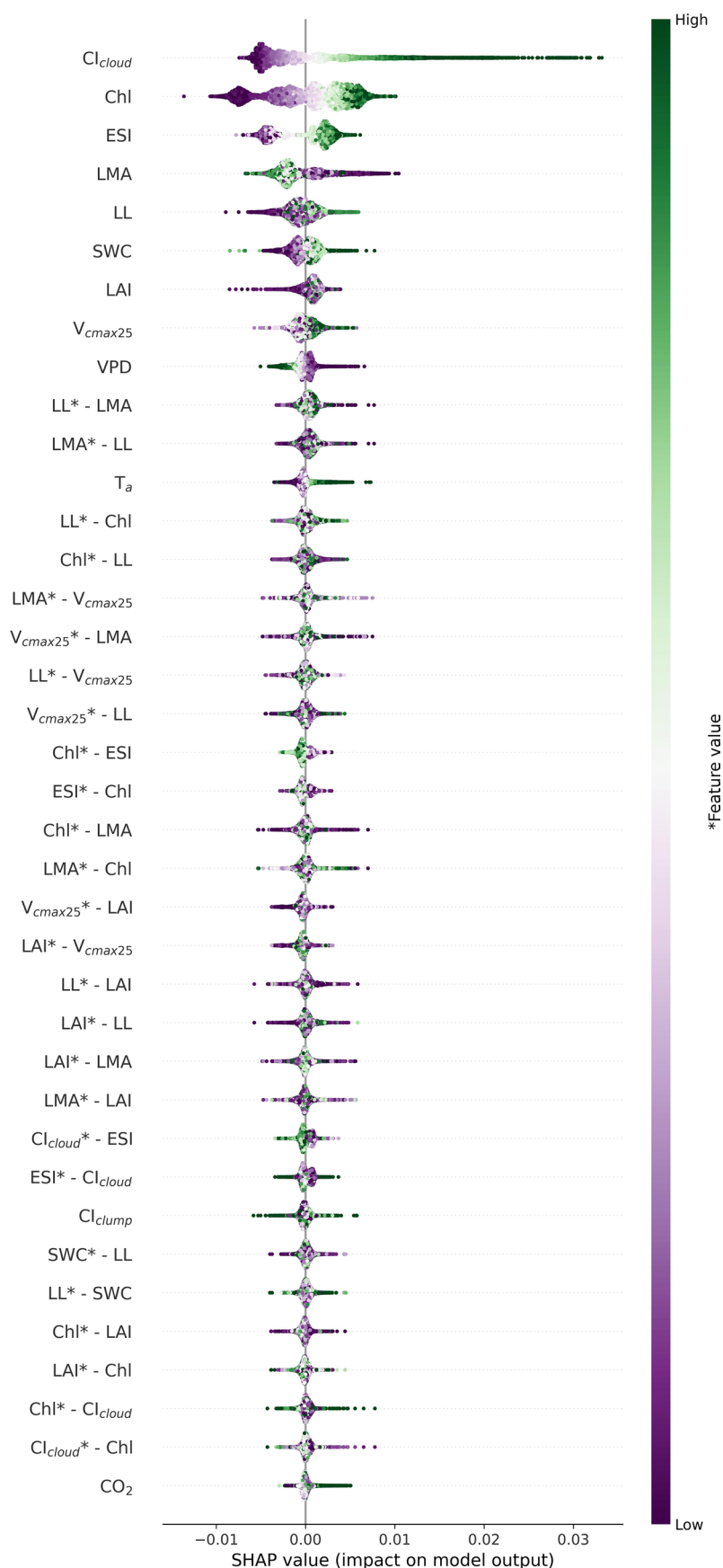
Extended Data Fig. 5 | Dominant (top six) features in predicting daily ecosystem PE across global major PFTs. a–j. Ranked in descending order of importance, along with their relative feature values. ESI: evaporative stress index;

Chl: leaf chlorophyll content; Cl_{cloud} : cloudiness index; LAI: leaf area index; LL: leaf longevity; LMA: leaf mass per area; SWC: soil water content; T_a : air temperature; V_{cmax25} : maximum carboxylation rate at 25 °C; VPD: vapor pressure deficit.

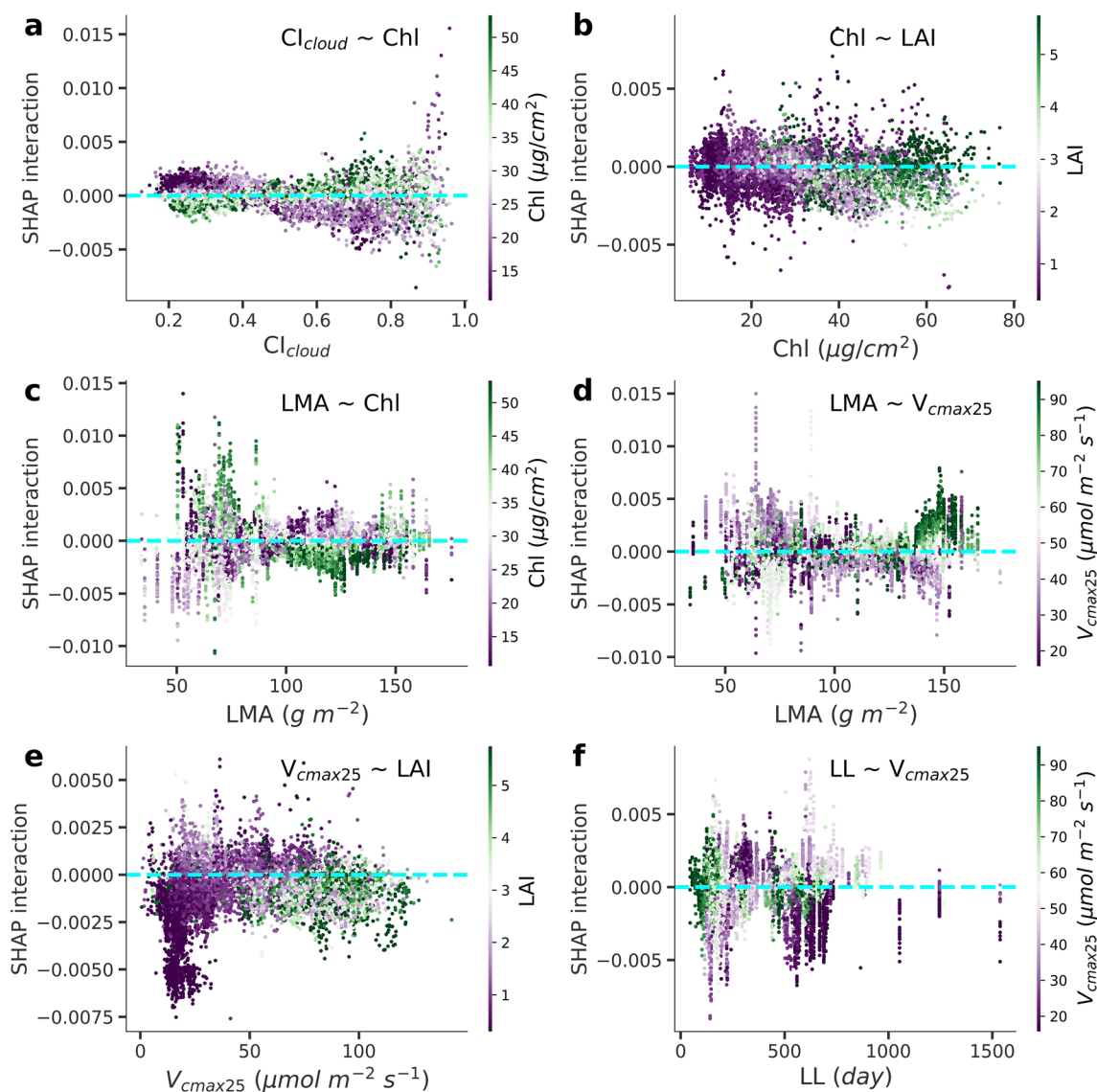


Extended Data Fig. 6 | Performance of the XGBoost model (trained and evaluated on a per-PFTs basis) in estimating daily ecosystem-scale PE across global major PFTs. a–j. CRO: cropland; CSH: closed shrubland; DBF: deciduous

broadleaf forest; EBF: evergreen broadleaf forest; ENF: evergreen needleleaf forest; GRA: grassland; MF: mixed forest; WSA: woody savanna; SAV: savanna; OSH: open shrubland. Shaded area indicates Gaussian kernel density estimation.



Extended Data Fig. 7 | List of feature's main effects and interaction terms on ecosystem PE. ESI: evaporative stress index; Chl: leaf chlorophyll content; Cl_{cloud} : cloudiness index; LAI: leaf area index; LL: leaf longevity; LMA: leaf mass per area; SWC: soil water content; T_a : air temperature; V_{cmax25} : maximum carboxylation rate at 25 °C; VPD: vapor pressure deficit.



Extended Data Fig. 8 | Feature interaction effects on daily ecosystem PE. **a–f**, SHAP interaction plots demonstrate how pairs of features interact to influence predictions. The vertical dispersion of the data points in the interaction figures indicate the presence and magnitude of interaction effects between the

variables. Specifically, a greater spread along the vertical axis indicates stronger interactions, while a narrower spread suggests weaker or negligible interaction. Cl_{cloud} : cloudiness index, Chl : leaf chlorophyll content; LAI : leaf area index; LL : leaf longevity; LMA : leaf mass per area; V_{cmax25} : maximum carboxylation rate at 25 °C.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- | | | |
|-------------------------------------|-------------------------------------|--|
| ☐ | ☒ | The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☒ | ☐ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☒ | ☐ | The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☒ | ☐ | A description of all covariates tested |
| ☒ | ☐ | A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ | For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ | Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection We did not use any commercial, open source, or custom code to collect data

Data analysis Relevant codes and package versions are available at <https://github.com/Heeheedaa/PE>

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The dataset used in this study can be accessed at the following link:
 FLUXNET2015: <https://fluxnet.org/data/fluxnet2015-dataset/>.
 MODIS LAI/fPAR: <https://modis.gsfc.nasa.gov/data/dataproduct/mod15.php>.
 GEOV2 fPAR: <https://land.copernicus.eu/en/products/vegetation>.

GLASS LAI: <http://www.glass.umd.edu/Download.html>.
 Clumping index: www.geodata.cn.
 MODIS chlorophyll content: <https://doi.org/10.5281/zenodo.5805575>.
 MERIS chlorophyll content: <https://zenodo.org/records/10467919>.
 Glopnet dataset: <https://doi.org/10.1038/nature02403>.
 The China Plant Trait Database: <https://doi.org/10.1002/ecy.2091>.
 The data supporting Figures 1–4 are available at <https://zenodo.org/records/14191212>.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	N/A
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	In this study, we integrated remote sensing and eco-evolutionary optimality theory to derive key plant traits, alongside explainable machine learning and global eddy covariance observations, to uncover the drivers of PE variations across global terrestrial ecosystems.
Research sample	We used FLUXNET2015 Tier 1 daily dataset, remotely sensed FPAR, LAI, Chl, and clumping index data. We perform spatial matching to align remotely sensed data with the footprint of the FLUXNET2015 dataset.
Sampling strategy	All the data available were used with the exception of data coming from wetland sites. No other sampling strategy was used.
Data collection	The data used in this study is publicly available. Links to data repositories are provided in the manuscript.
Timing and spatial scale	Daily net ecosystem exchange and meteorological records measured at FLUXNET2015 eddy covariance sites. Daily, 4-day, or 8-day remotely sensed FPAR, LAI, Chl, and clumping index data.
Data exclusions	FLUXNET2015 Tier 1 data were filtered using a data quality control threshold set at >0.8. FPAR and LAI data were also quality controlled (e.g., FparLai_QC & FparExtra_QC). More information described in Method section.
Reproducibility	The data used in this study is publicly available. All code to analyze data are available on GitHub.
Randomization	To ensure reliable estimates of machine learning models performance, we implemented a k-fold cross-validation scheme with k set to either five or ten. In each iteration, 80% of the data was utilized for training, while the remaining 20% was allocated for validation.
Blinding	Blinding is not relevant to our study. The role of leaf traits in regulating leaf-level photosynthesis is well-established. Our focus is on employing explainable machine learning models and global datasets to determine the extent to which plant traits account for the substantial unexplained variability in ecosystem-scale photosynthetic efficiency.
Did the study involve field work?	☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A